



302 TXOne solutions troubleshooting

Version 1.5

Objective



Agenda

1. Troubleshooting Network Defense Solutions (Edge)
2. Troubleshooting Endpoint Protection Solutions (Stellar)
3. Troubleshooting Security Inspection Solutions (Element)



Troubleshooting Network Defense Solutions

Scenarios

1. The EdgeOne / ODC not booting
2. Unable to access EdgeOne web UI
3. Can not update firmware / patterns
4. Unable to sync the EdgeIPS to the EdgeOne / ODC
5. Auto rule learning doesn't learn any rules
6. Can not activate license
7. Forgot logon credential
8. Debug log collection
9. What are the steps to analyze hardware failure?
10. How to migrate from ODC to EdgeOne

Scenario 1 - The EdgeOne / ODC unable to boot up

Requirement

EdgeOne



- Image corrupt
- VMWare
 - ESXi6
 - Workstation 14
- Manage up to 200 node
 - CPU: 8 cores
 - RAM: 32GB

Forgot to create 2nd disk

EdgeOne



Disk1



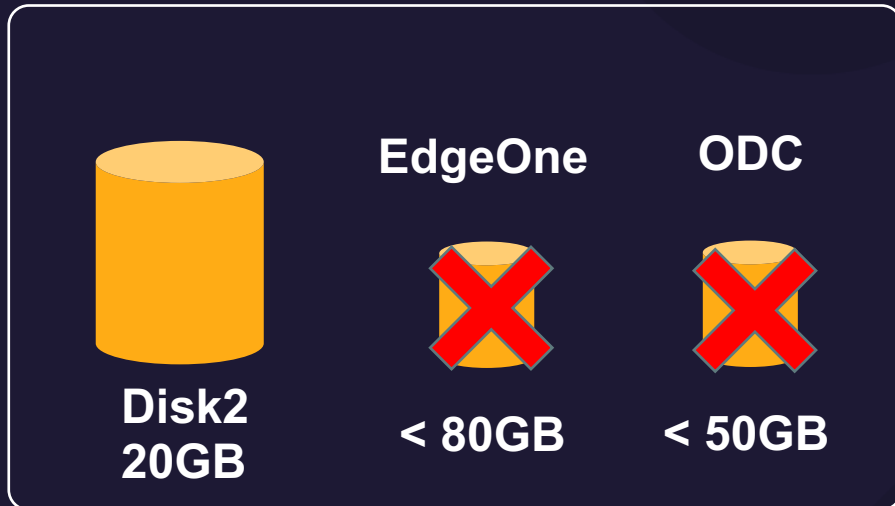
Disk2

- Logon EdgeOne server (admin / P@ssw0rd)
- **env ls** (Check the EdgeOne status)

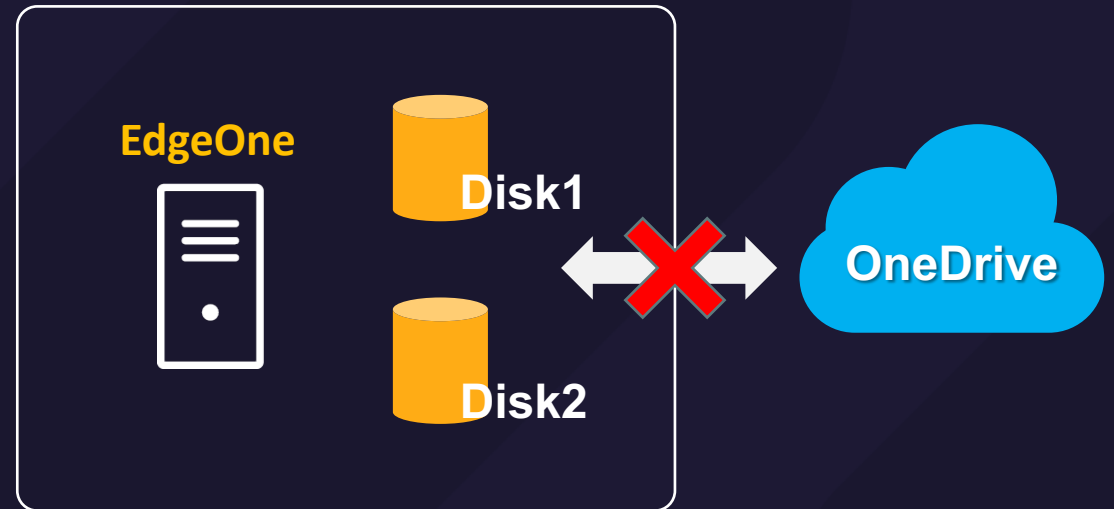
```
$ env ls
Hostname: EdgeOne
Status: FAIL: NO EXTERNAL DISK DETECTED
Product Serial Number:
Version:
```

Scenario 1 - The EdgeOne / ODC not boot up

The size of the 2nd disk smaller than 80GB / 50GB

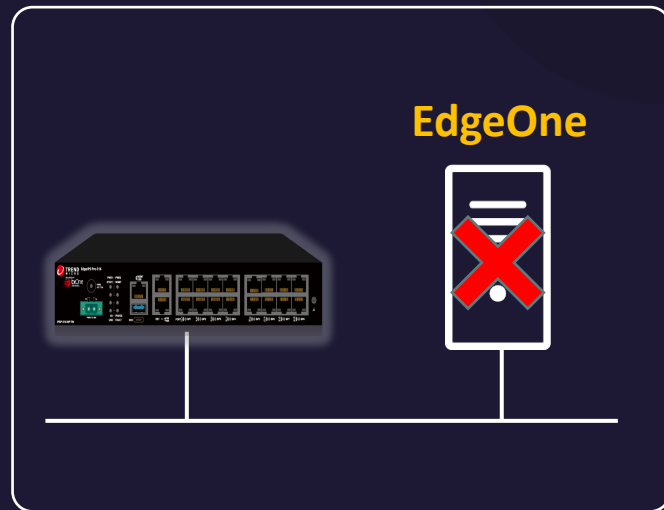


The Disk sync with Cloud drive services



Scenario 2 – Unable to access EdgeOne web UI

Network connection issue



- The firewall denies access from client to EdgeOne

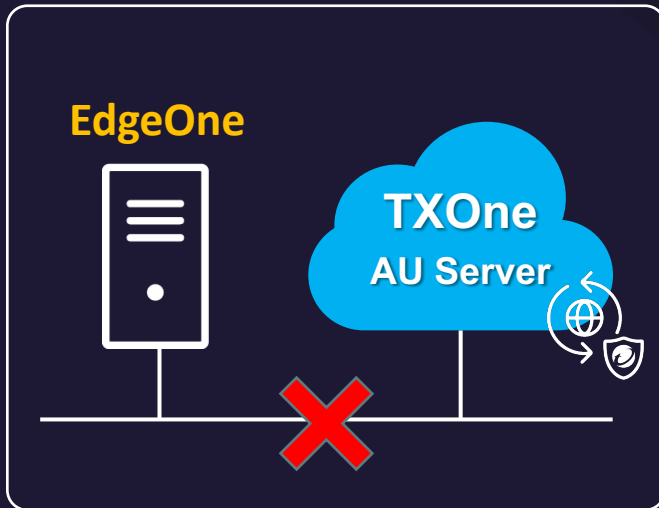
Browser issue



- The browser not use Chrome

Scenario 3 - Can not update firmware / patterns

Network connection issue



- The firewall deny the access to TXOne AU servers
- Network routing issue

License issue

The diagram shows a table with license information. A red 'X' is placed over the 'EdgeOne Node License' entry, indicating an issue with the license.

License Type	Seat
EdgeOne Node License	40

The Edge device license not activated

The pattern file corrupted For local import



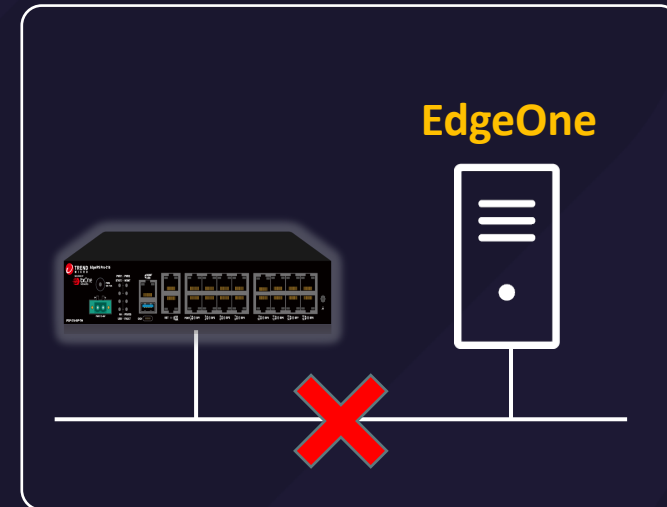
Scenario 4 – Unable to sync the EdgeIPS to the EdgeOne / ODC

Time sync issue



The timestamps not in sync between EdgeOne and Edge devices

Network connection issue



- The firewall denies access from Edge devices to EdgeOne
- EdgeOne behind NAT

Scenario 4 – Unable to sync the EdgeIPS to the EdgeOne / ODC

License issue



License Type	Seat
EdgeOne Node License	2
EdgeFire Software License	2
EdgeIPS 102 / 103 Software License	40

- Device seat count > Node license
- The Edge device license not activated

The wrong passcode



EdgeOne Server Address* 192.168.23.21

Authentication Method Passcode ▼

The wrong passcode

Test Connection

Scenario 5 - Auto rule learning doesn't learn any rules

The IP, ports which with in Enforcement ruleset

Policy Settings

IP: any
Port: any



Traffic Learned

192.168.23.100
tcp 80, 22

Click on the “Cancel the task” button

Auto-Learning Status	In Progress
IPS Profile	Default_IPS_Profil
IPS Mode	Monitor Mode
Period	2023-07-05T14:46

Stop the Task



Cancel the Task

Scenario 6 - Can not activate license

The wrong license key

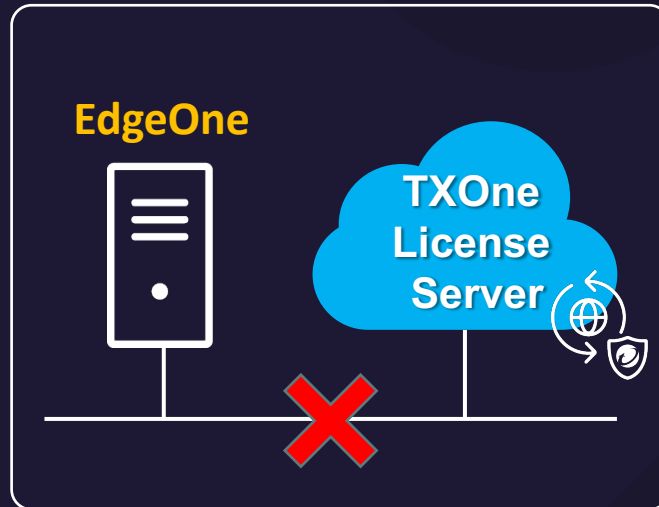


The license expired



Scenario 6 - Can not activate license

Network connection issue

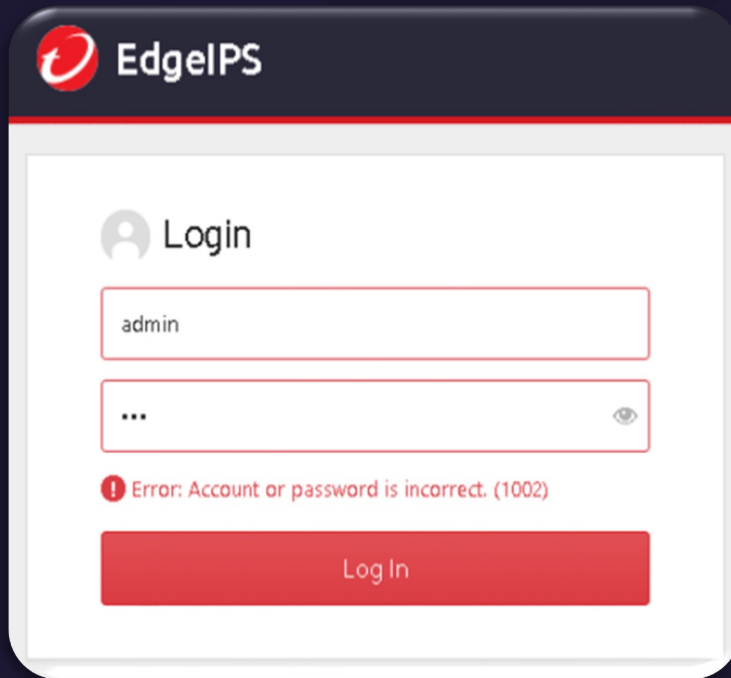


The license file corrupted for offline activation



Scenario 7 – Forgot logon credential

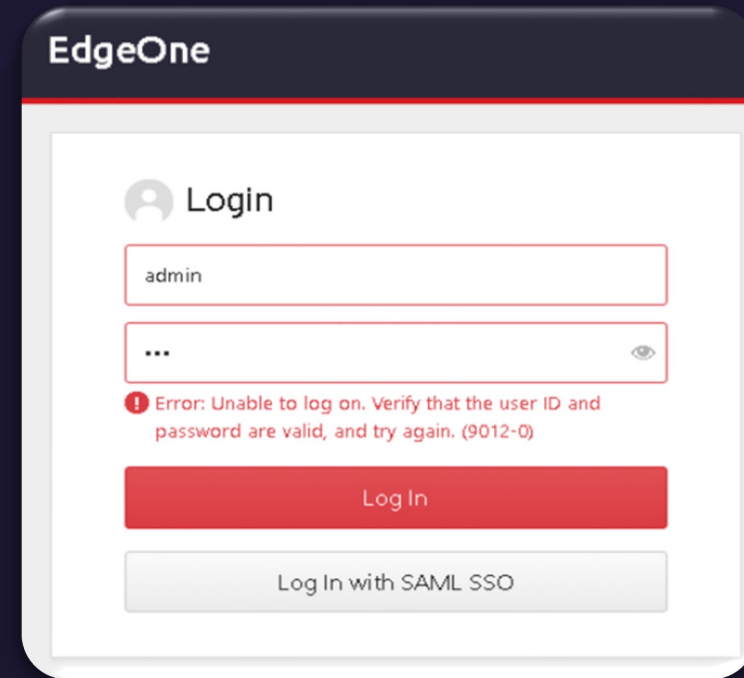
Edge devices



The EdgeIPS login interface features a dark header with the EdgeIPS logo. Below the header, a 'Login' section contains two input fields: the first is labeled 'admin' and the second is masked with three dots. A red error message is displayed below the password field: 'Error: Account or password is incorrect. (1002)'. At the bottom, there is a red 'Log In' button.

- Hardware factory reset

EdgeOne

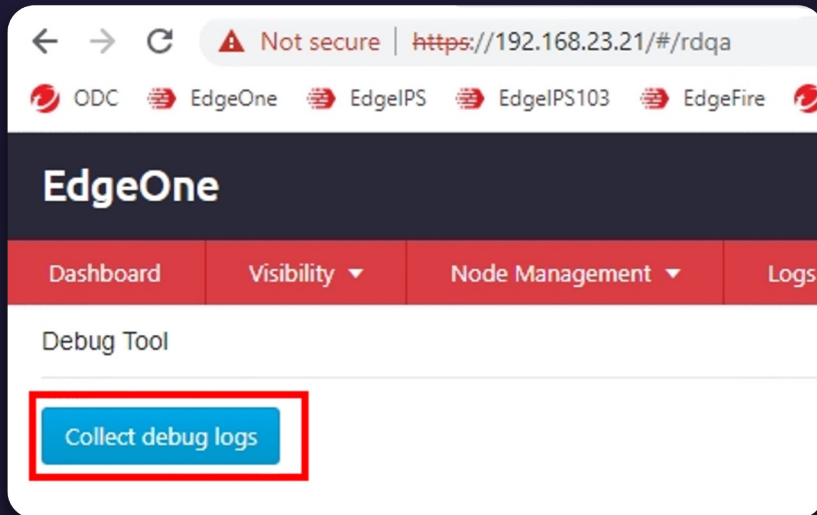


The EdgeOne login interface has a dark header with the EdgeOne logo. The 'Login' section includes two input fields: the first contains 'admin' and the second is masked with three dots. A red error message is shown below the password field: 'Error: Unable to log on. Verify that the user ID and password are valid, and try again. (9012-0)'. Below the error message are two buttons: a red 'Log In' button and a grey 'Log In with SAML SSO' button.

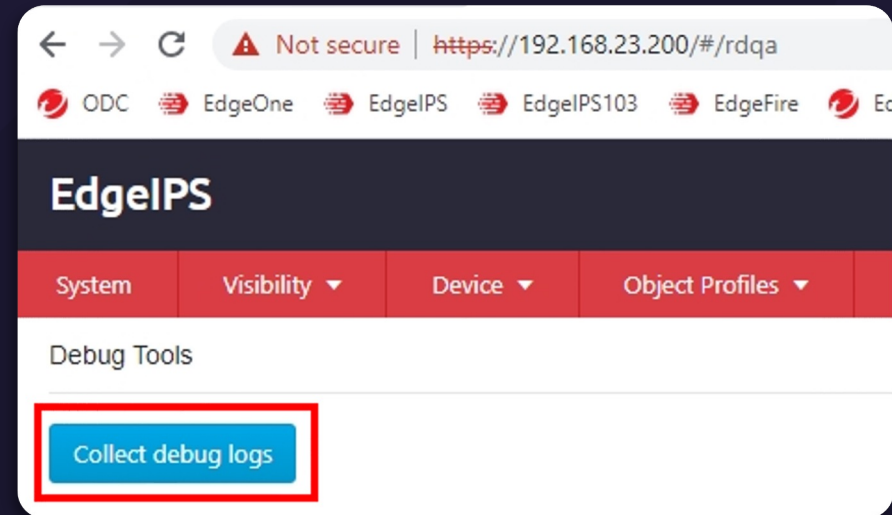
- Vshell : web reset admin

Scenario 8 - Debug log collection

- EdgeOne
 - https://EdgeOne_web_ui/#/rdqa



- Edge devices
 - https://Edge_device_ip/#/rdqa

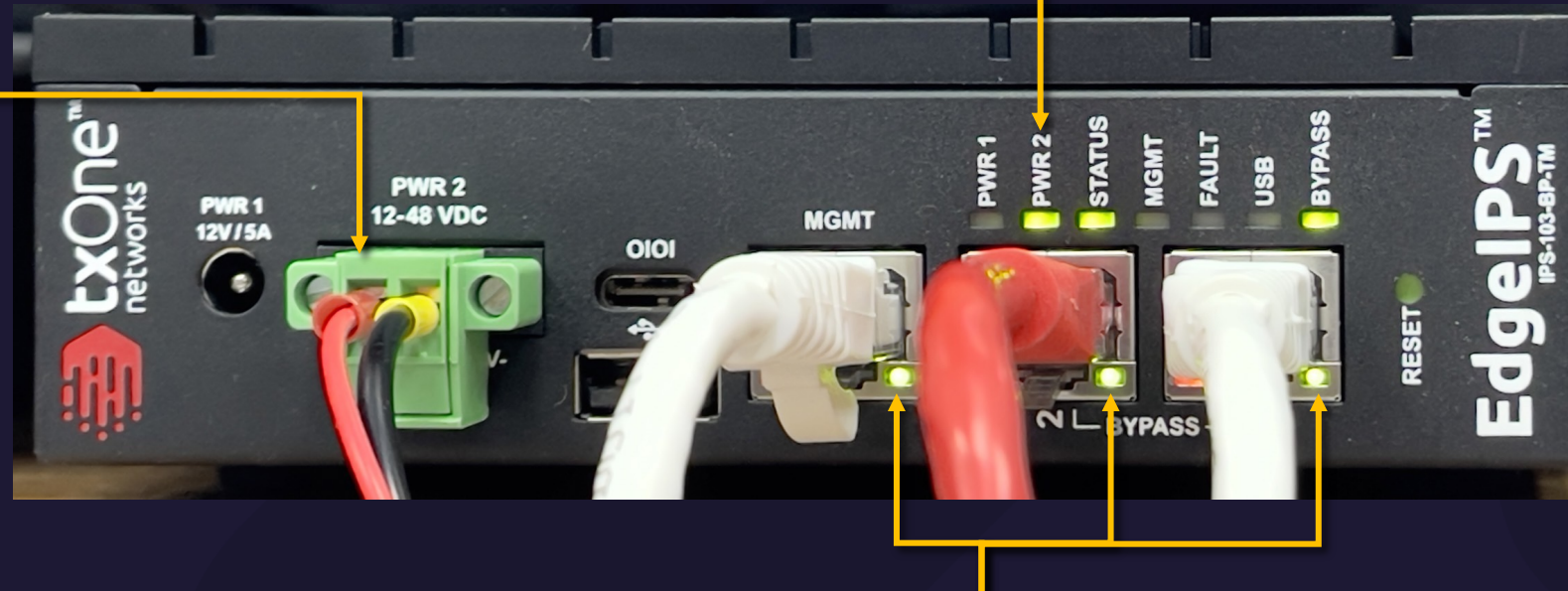


Scenario 9 - What are the steps to analyze hardware failure?

- Check if power adapter or cable is connected to devices without issues

Checking power adapter connected and PWR1 or PWR2 with a solid green LED

check the power v+ v- connect and power supply is ok.

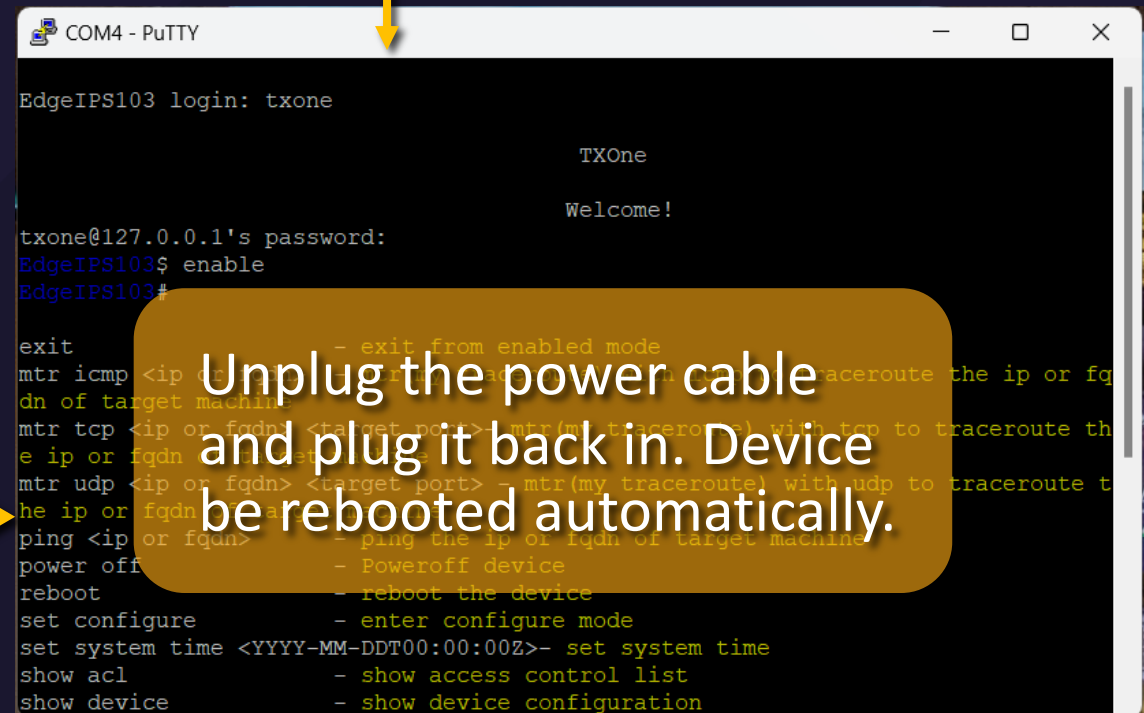
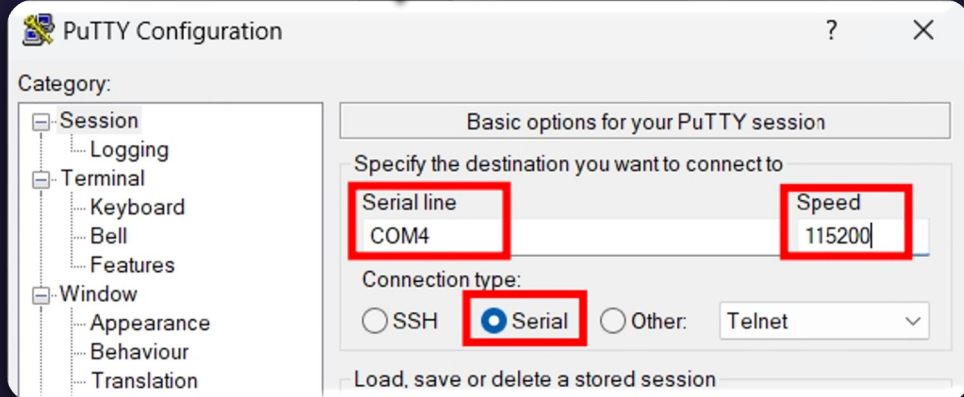
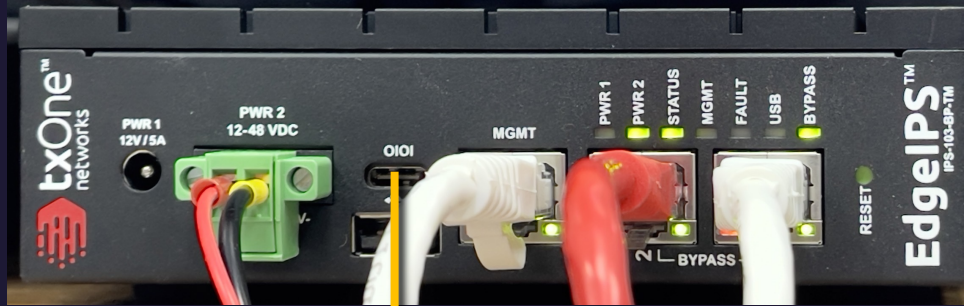


Verify the network cable is properly connected and make sure the LAN port LED next to the network jack are properly illuminated

Scenario 9 - What are the steps to analyze hardware failure?

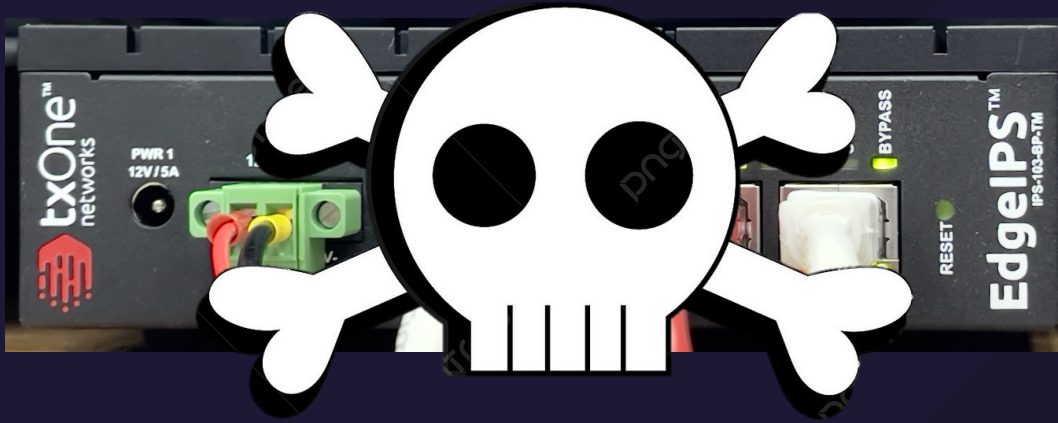
Identifying the hardware issue

- Edge device system hang: If there is **no output** on the serial console, such as no response to input on the serial console.



Scenario 9 - What are the steps to analyze hardware failure?

Identifying the hardware issue



Try to Factory Reset
to default settings

- Edge device system crash
 - The Edge device product experiences a reboot or reload while either doing a specific action or randomly
- System Crash and boot loop
 - The Edge device product can be stuck in the continuous loop with an error message scrolling.

Scenario 10: How to migrate from ODC to EdgeOne

Preparation

1 ODC v1.4 or above



ODC*



ODC License Key



ODC Configuration File



ODC 2nd Disk
(Optional)

*Please update the pattern version to the latest version before backing up the configuration file.

2 New VM Environment for EdgeOne



Launch New VM*

*Please refer to the system requirement to prepare the new VM.

Scenario 10: How to migrate from ODC to EdgeOne

Migration Flow for ODC & Edge Series

Mandatory

System

Configuration

License

1 Back up the ODC configuration file

ODC-VA
V1.X

2 Copy the license keys

ODC-VA
V1.X

3 Initial a new VM environment (New IP Address)

EdgeOne
V2.2

4 Activate the license on EdgeOne

EdgeOne
V2.2

5 Restore the EdgeOne configuration from the ODC configuration file

EdgeOne
V2.2

6 Power off ODC

ODC-VA
V1.X

7 Update EdgeOne IP Settings to ODC's IP

EdgeOne
V2.2

Optional

Legacy Logs

Power off EdgeOne and then attach ODC second disk as EdgeOne's third disk (Optional)

***It is not necessary to migrate the patterns since the V1 and V2 patterns are completely different.**

B. Migrate from Edge V1.X to Edge V2



Edge V1.X firmware

Edge V2 firmware

Scenario 10: How to migrate from ODC to EdgeOne

Legacy Log Viewer

- When added ODC disk as 3rd disk to EdgeOne (2.2 or later version) successfully, show '**Legacy Logs**' under Logs.
- The log table columns will follow the ODC 1.5.x spec definition.

EdgeOne

admin (Admin)

Dashboard Visibility Node Management Logs Report Application Administration About

Logs > Legacy Logs

Cyber Security Logs Latest 5000 records 2021-03-29T09:13:12+08:00 to 2022-03-29T09:13:12+08:00 Add Filter(s) Search

Time Range: 2021-06-01T16:39:40+08:00 to 2022-03-29T09:13:12+08:00

Time	Device Name	Serial Number	Function Type	Protocol Layer	Rule Name	Profile Name	Event ID	TID	Security Category	Security Severity
2022-06-03T16:38:57+08:00	IPS	TMF01-3b38ac180022	System Security	Layer2	Device:Rule1	Device:Profile1	4026531846	T1234.003;	Flooding & Scan	High
2022-06-03T16:38:57+08:00	IPS								Flooding & Scan	High
2022-06-03T16:38:57+08:00	IPS								Flooding & Scan	High
2022-06-03T16:38:57+08:00	IPS								Flooding & Scan	High
2022-06-03T16:38:57+08:00	IPS								Flooding & Scan	High
2022-06-03T16:38:57+08:00	IPS								Flooding & Scan	High
2022-06-03T16:38:57+08:00	IPS	TMF01-3b38ac180022	System Security	Layer2	Device:Rule1	Device:Profile1	4026531846	-	Flooding & Scan	High
2022-06-03T16:38:57+08:00	IPS	TMF01-3b38ac180022	Cyber Security	Layer3	EdgeOne:Rule2	EdgeOne:Profile2	4026531846	T1234.003;	Flooding & Scan	High

Permission:

- All legacy logs: Master Admin, System Admin
- Legacy Audit Logs: Auditor

Cyber Security Logs

- Policy Enforcement Logs
- Protocol Filter Logs
- File Filter & Antivirus Logs
- Suspicious Object Logs
- Asset Detection Logs
- System Logs
- Audit Logs

Endpoint Protection Solutions

Troubleshooting Scenarios

1. The StellarOne unable to boot
2. The StellarOne pattern update issues
3. Agents not showing in the StellarOne web UI
4. Agent installation, failed to repack the agent installer
5. How to do if forget vshell console's root password?
6. Stellar agents fail to install due to incompatible Windows Defender
7. StellarProtect agents fail to install due to the .NET framework missing
8. Debug log collection

see also: <https://help.txone.com/v1/docs/troubleshoot-3>

Scenario 1 - The StellarOne unable to boot up

Requirement

StellarOne



- Image corrupt
- VMWare
 - ESXi6
 - Workstation 14
- Manage up to 20,000 node
 - CPU: 8 cores
 - RAM: 16 GB

Forgot to create 2nd disk

StellarOne



Disk1



Disk2

- Logon StellarOne server (root / P@ssw0rd)
- **env ls** (Check the StellarOne status)

If you want to exit this shell, please type `exit` or `Ctrl-D`.

```
$ env ls
```

```
Hostname: nnc
```

```
Status: FAIL: NO EXTERNAL DISK DETECTED
```

```
Product Serial Number: d1368284-2cf8-11ed-ae40-000c29edfc47
```

```
Version: 1.5.8
```

```
External IP: Not Set
```

```
DPI Engine Version: 2.0.11.33e2e1
```

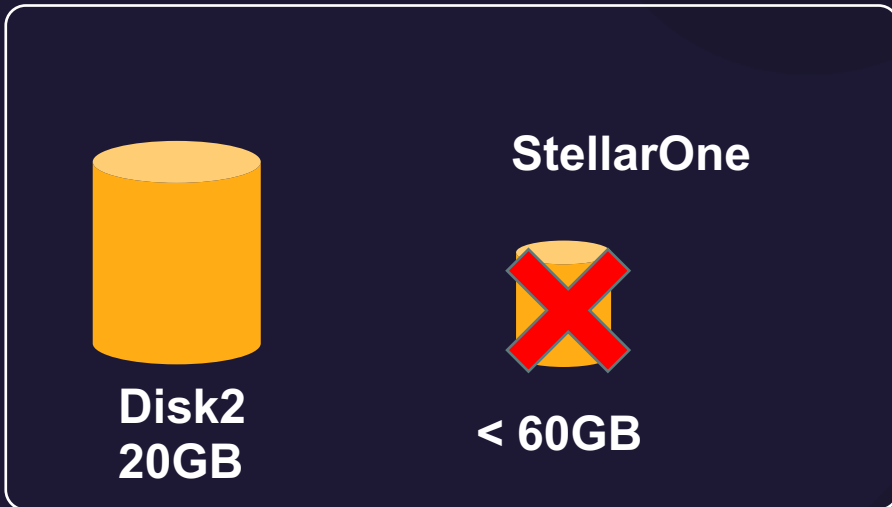
```
DPI Pattern Version: SDP_211215_12
```

```
Service Network: config: 169.254.0.0/16, network: [169.254.0.0/24]
```

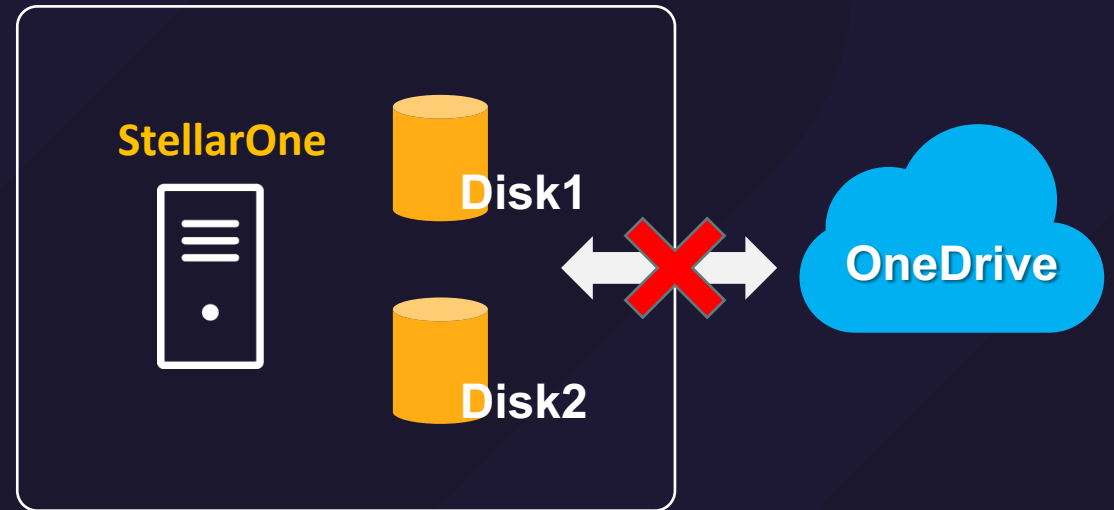
```
$
```

Scenario 1 - The StellarOne not boot up

The size of the 2nd disk smaller than 60GB

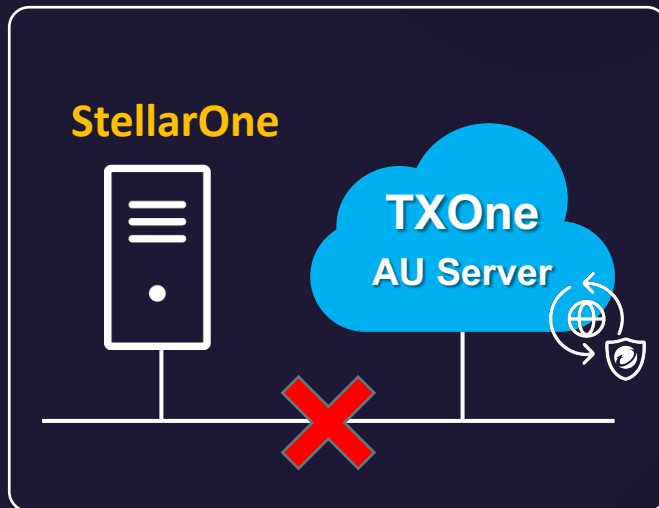


The Disk sync with Cloud drive services



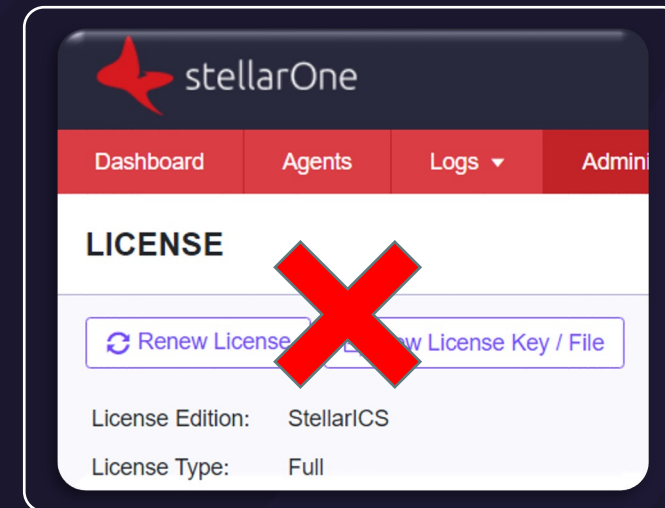
Scenario 2 - StellarOne pattern update issues

Network connection issue



- The firewall deny the access to TXOne AU servers
- Network routing issue

License issue



- The license expired
- The wrong license key

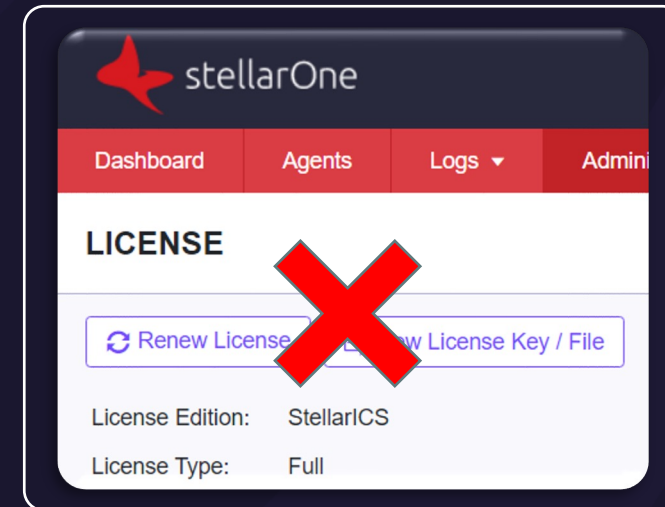
Scenario 3 - Agents not showing in the StellarOne web UI

Time sync issue



The timestamps not in sync between StellarOne and Agents

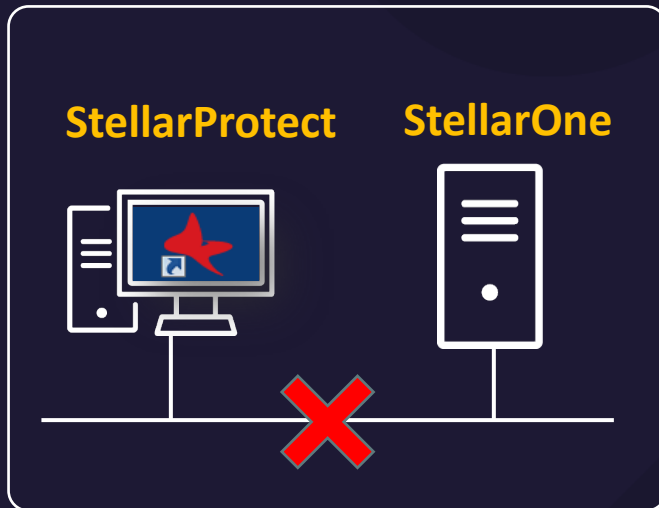
License issue



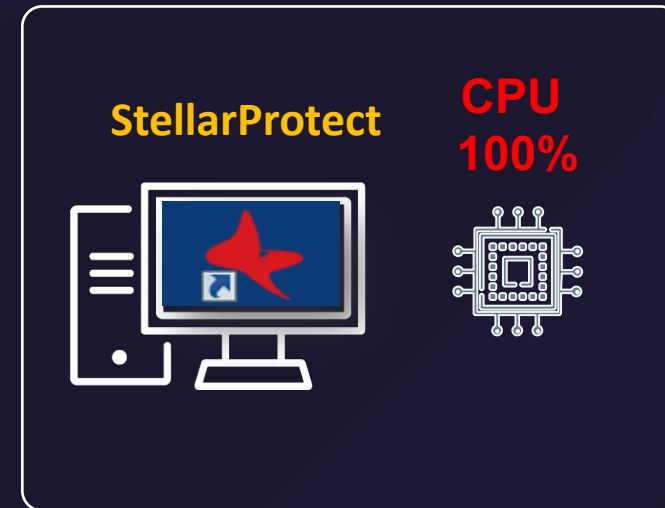
- The license expired
- The wrong license key

Scenario 3 - Agents not showing in the StellarOne web UI

Network connection issue



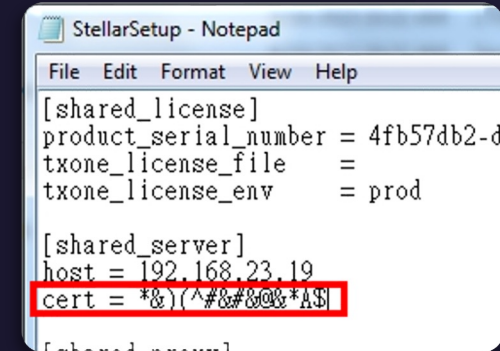
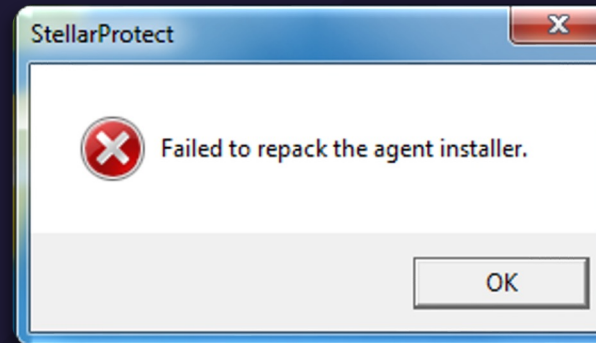
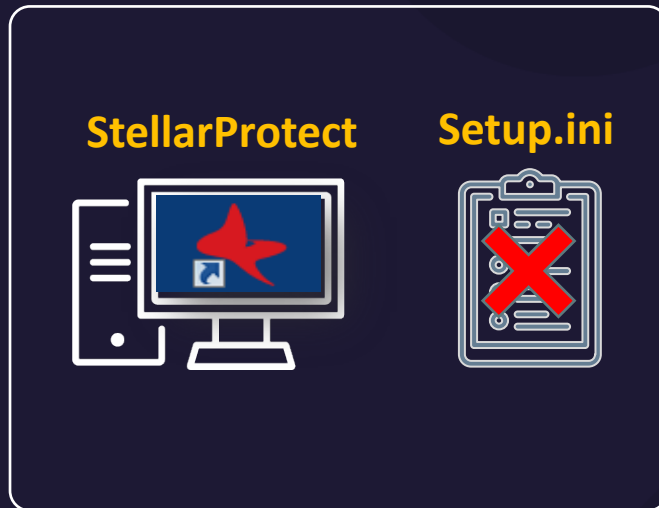
Client OS resources exhausted



- Firewall denies access from agents to StellarOne
- StellarOne behind NAT
- Proxy in path but not configured in Agent installer package (StellarProtect.ini)

Scenario 4 – Agent installation, failed to repack the agent installer

StellarSetup.ini format error

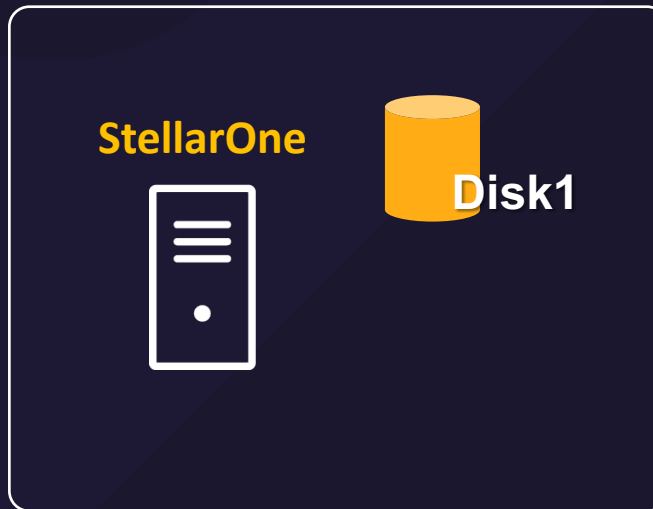


Scenario 5 – How to do if forget vshell console's root password?

Backup StellarOne disk image included Disk 1 & 2



Install new StellarOne and assign same IP address



Logon on vshell with default password root / txone

```
ODC login: root
Password:
Linux ODC 5.10.0-8-amd64 #1 SMP De

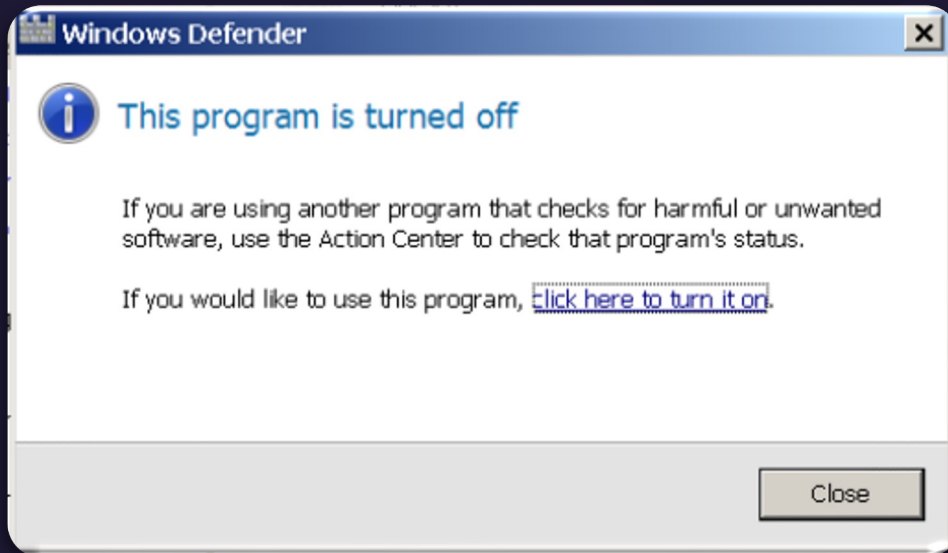
The programs included with the Deb
the exact distribution terms for e
individual files in /usr/share/doc

Debian GNU/Linux comes with ABSOLU
permitted by applicable law.
vShell, version
```

Scenario 6: Stellar agents fail to install due to incompatible Windows Defender

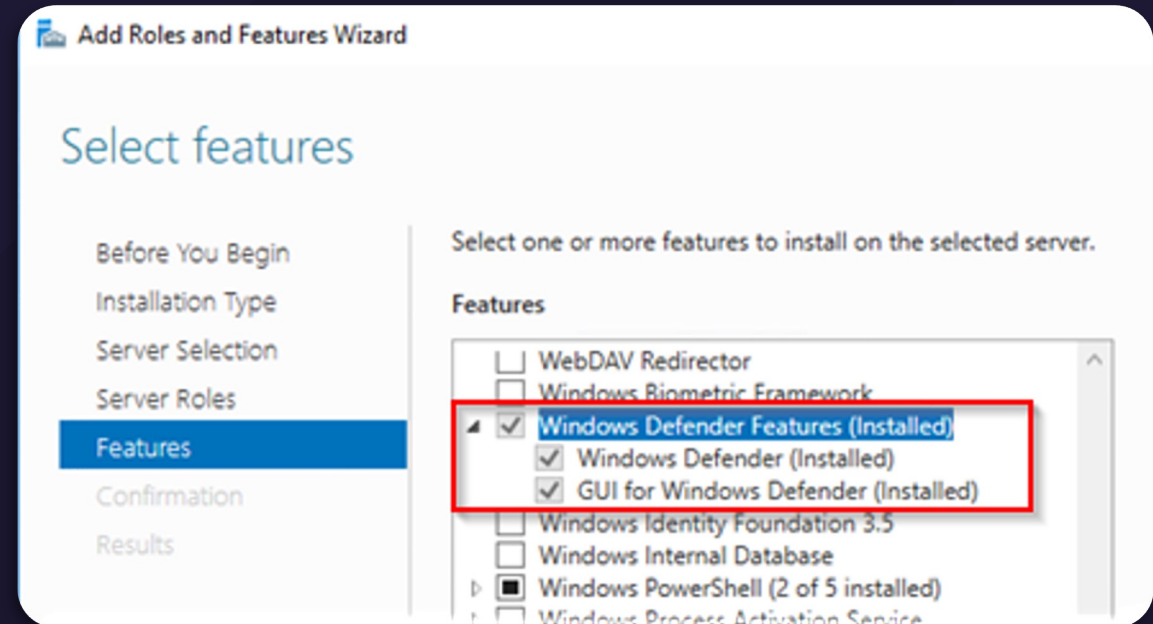
Windows client

Turn off Windows Defender



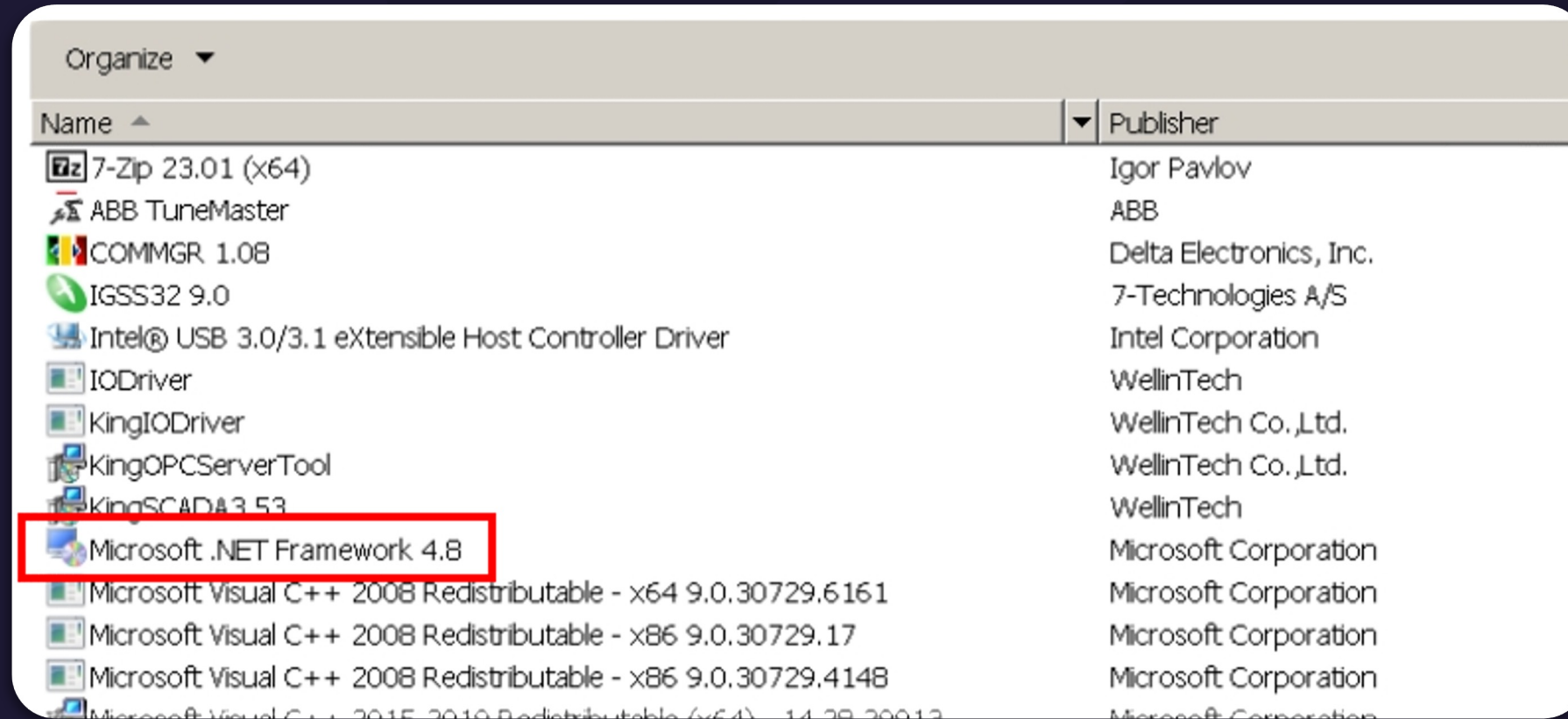
Windows Server

Remove Windows Defender feature



Scenario 7: StellarProtect agents fail to install due to the .NET framework missing

- Reinstall .NET Framework package



Scenario 7: Stellar Debug Log Collection when installation fails

Stellar v3.0 SP1 and newer

StellarProtect / StellarProtect Legacy Mode installation fails

- Modify StellarSetup.ini to enable debug log collection and try the installation again.

Default Value

```
[shared_proxy]
host =
port =
username =
password =

[shared_install]
silent = 0
password =
user_password =

[shared_debug]
enable_debug_log = 0
enable_engine_debug_log = 0

[protect_server]
port = 9443

[protect_listen]
port = 14336
```

Parameters and values

enable_debug_log: 0 (error only log) | 1 (verbose log)

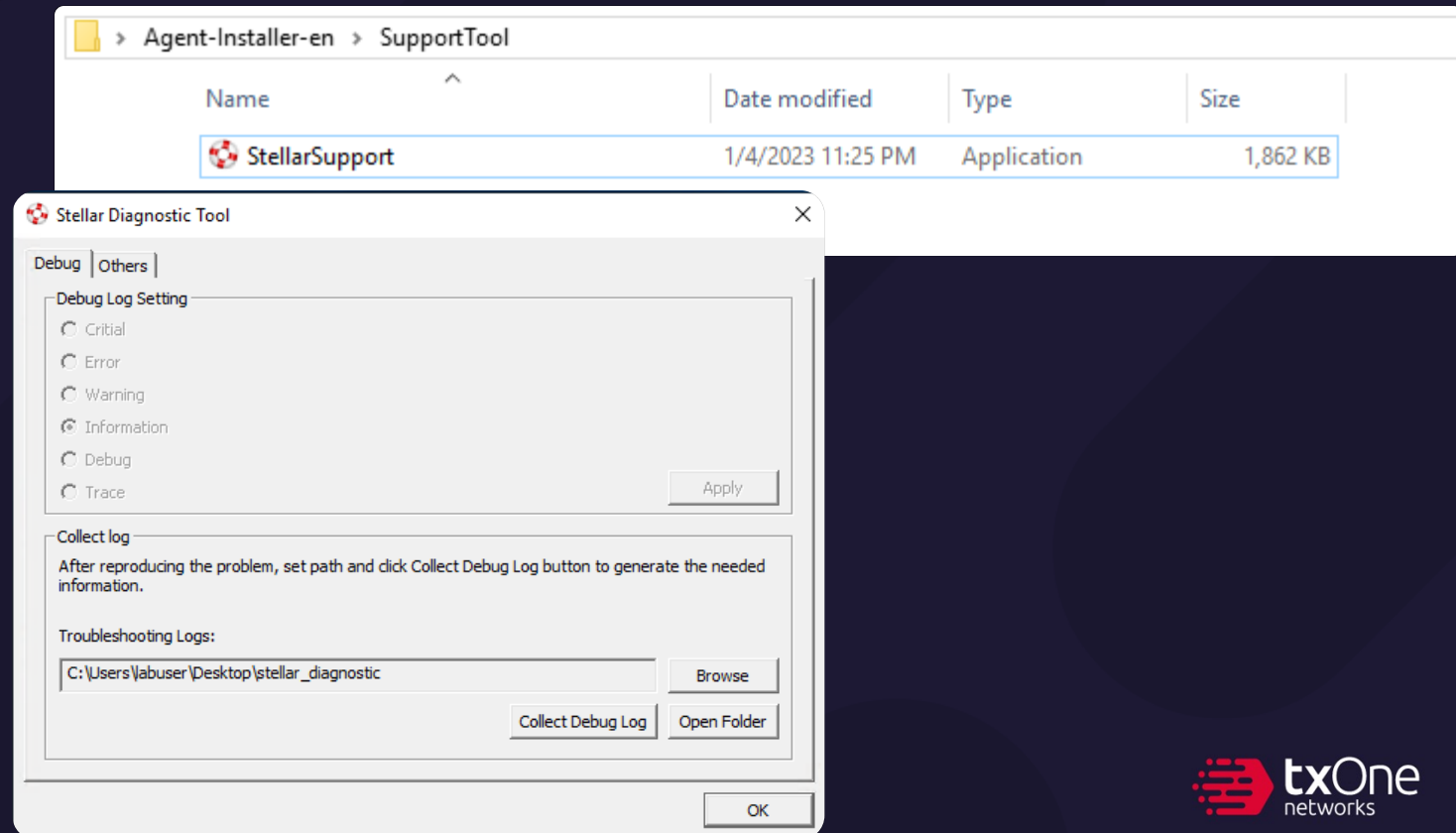
enable_engine_debug_log: 0 (no log) | 1 (verbose log)

enable_debug_log: 1
enable_engine_debug_log: 1

Please only modify these two debug settings for troubleshooting.
DO NOT MODIFY in normal installation to avoid performance issues

StellarProtect / StellarProtect Legacy Mode installation fails

- Collect the StellarProtect logs using StellarSupport.exe
 - Modify the StellarSetup.ini (see previous slide)
 - Re-run the installer to reproduce the issue, and do not close any installation windows the installation fails
 - Execute StellarSupport.exe (in the Agent Installer Package)
 - Click “Collect Debug Log”
 - Click “Open Folder” to get the archived zip file

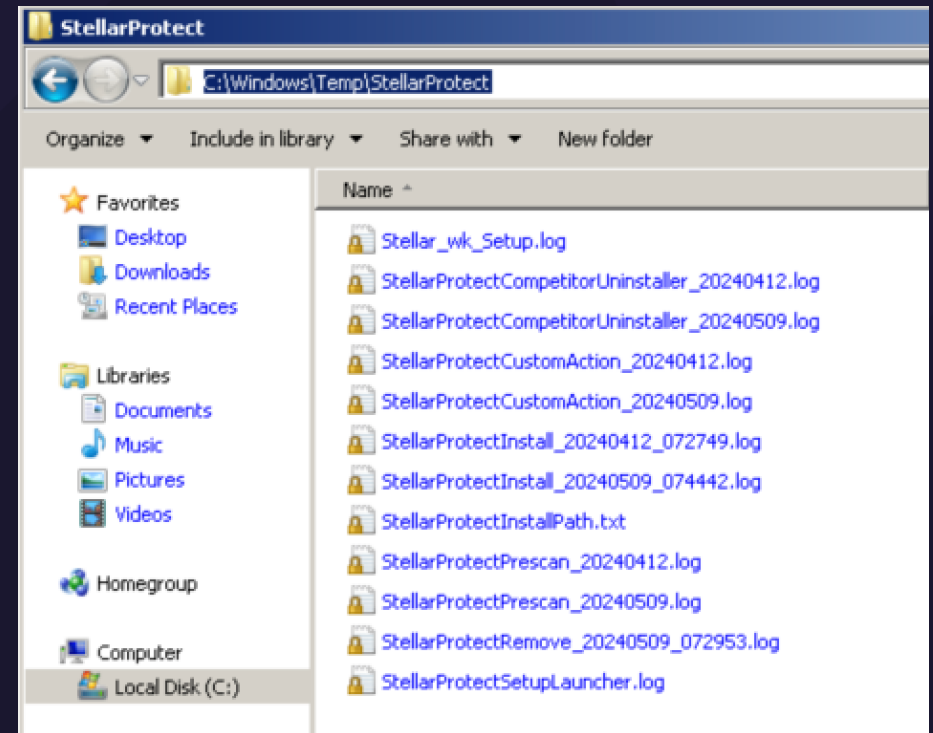


StellarProtect / StellarProtect Legacy Mode installation fails

- Locally, the agent-installation-log files can be found under:
`c:\Windows\Temp\StellarProtect`

- Pro tip:
to find out in which log file to look for errors,
run the following powershell command:

```
Get-ChildItem "C:\Windows\Temp\StellarProtect" -Filter  
 "*.log" -Recurse | Select-String "error"
```

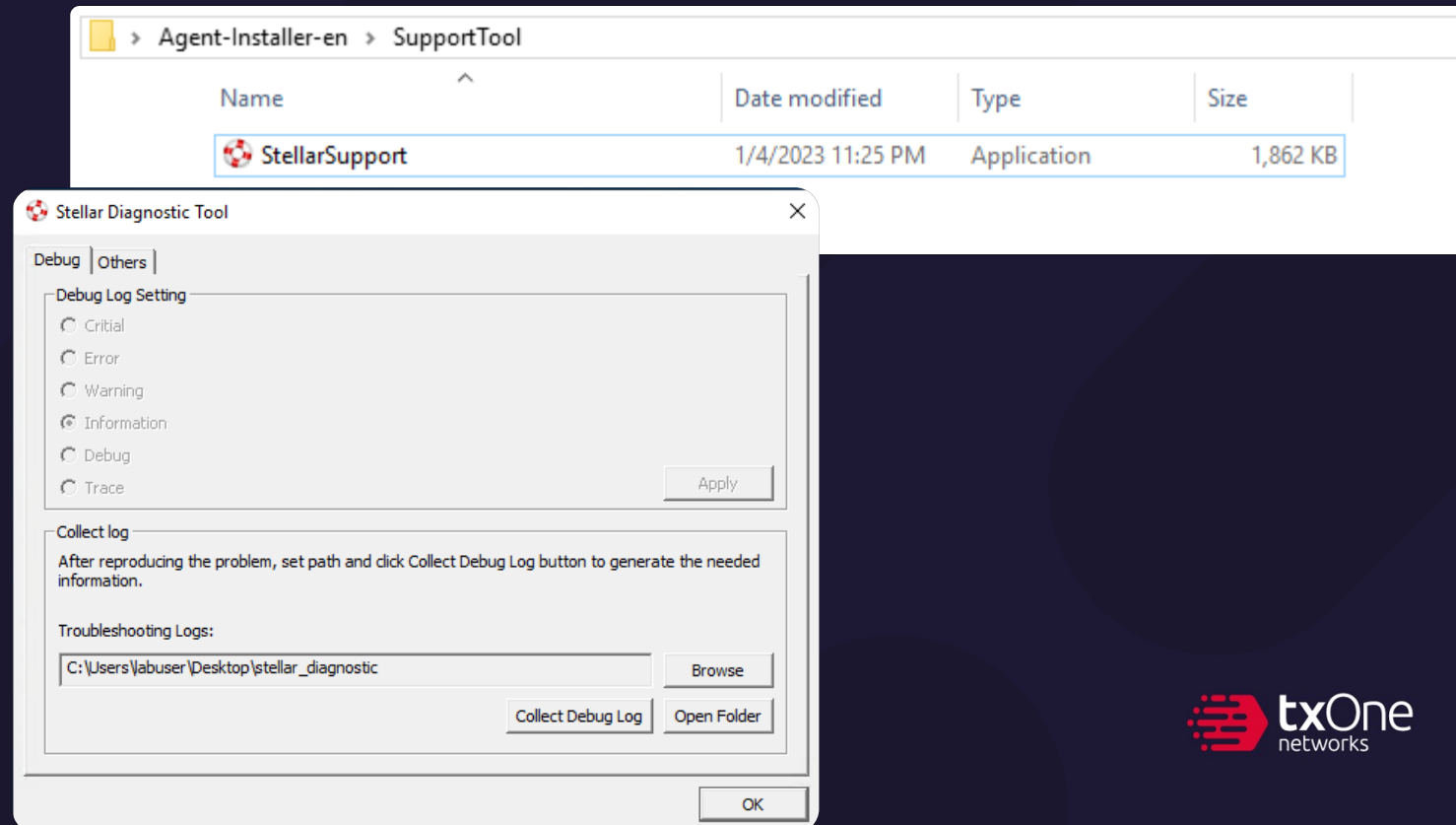


Scenario 7: Stellar Debug Log Collection when installation fails

Before Stellar v3.0 SP1

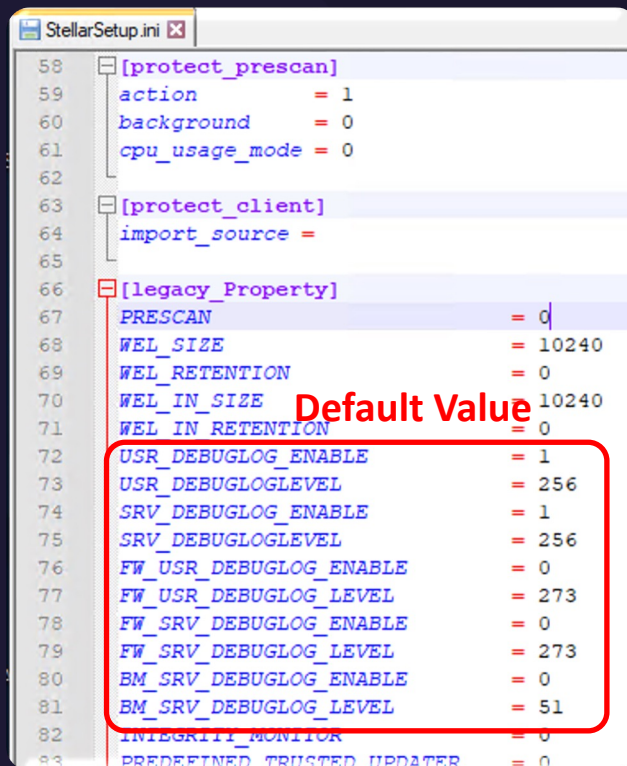
StellarProtect installation fails

- In older versions of StellarProtect, the Support tool is different for SP vs. SPLM
- Collect the StellarProtect logs using StellarSupport.exe
 - *For StellarProtect, older versions, there are no settings for debug log in StellarSetup.ini*
 - Re-run the installer to reproduce the issue, and do not close any installation windows when the installation fails
 - Execute StellarSupport.exe (in the Agent Installer Package)
 - Click “Collect Debug Log”
 - Click “Open Folder” to get the archived zip file



StellarProtect *Legacy* Mode installation fails

- Increase the debug log level collection settings in StellarSetup.ini (which is in the agent installation package folder)



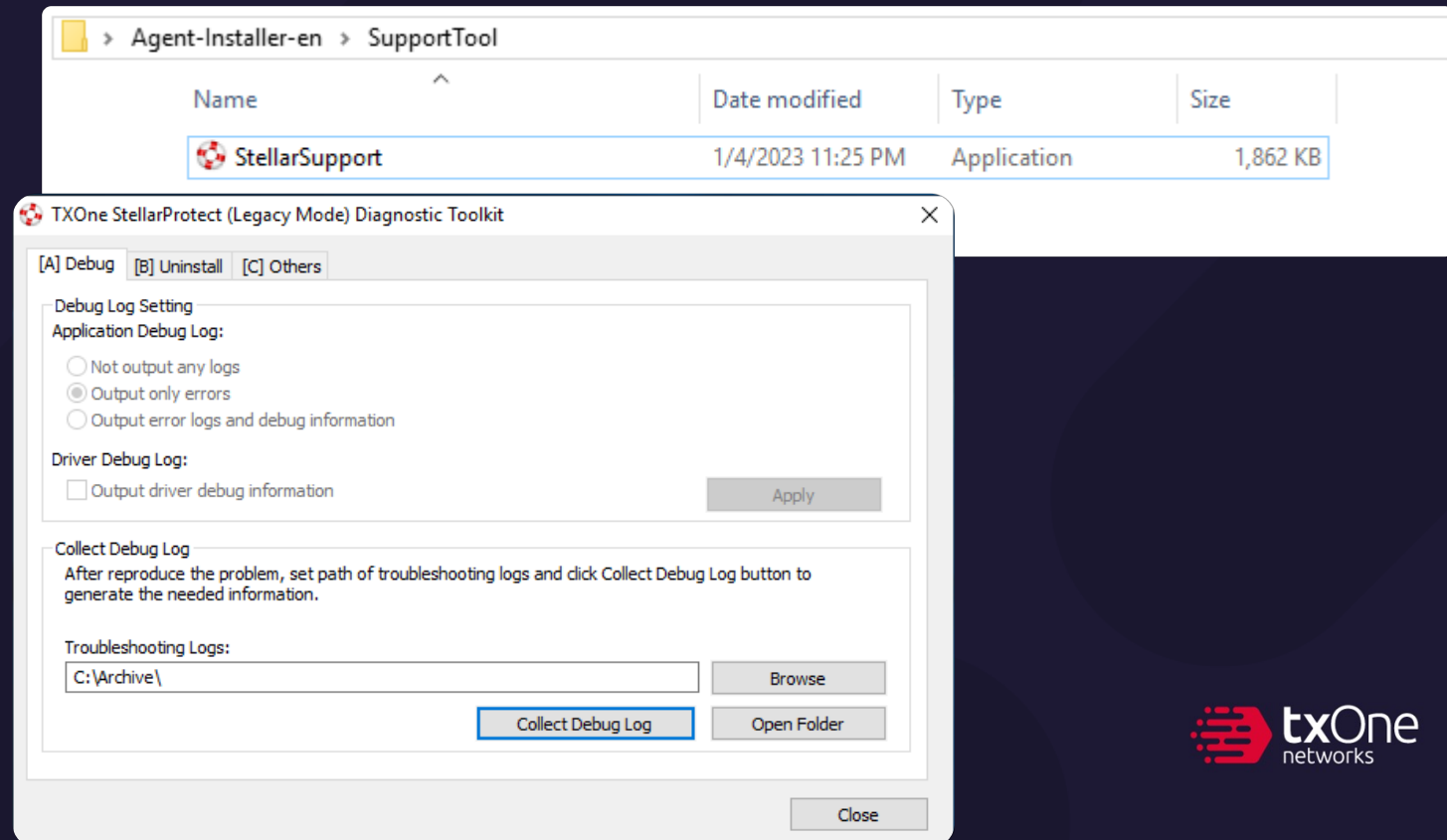
```
StellarSetup.ini
58 [protect_prescan]
59   action      = 1
60   background  = 0
61   cpu_usage_mode = 0
62
63 [protect_client]
64   import_source =
65
66 [legacy_Property]
67   PRESCAN      = 0
68   WEL_SIZE     = 10240
69   WEL_RETENTION = 0
70   WEL_IN_SIZE  = 10240
71   WEL_IN_RETENTION = 0
72   USR_DEBUGLOG_ENABLE = 1
73   USR_DEBUGLOGLEVEL   = 256
74   SRV_DEBUGLOG_ENABLE = 1
75   SRV_DEBUGLOGLEVEL   = 256
76   FW_USR_DEBUGLOG_ENABLE = 0
77   FW_USR_DEBUGLOG_LEVEL  = 273
78   FW_SRV_DEBUGLOG_ENABLE = 0
79   FW_SRV_DEBUGLOG_LEVEL  = 273
80   BM_SRV_DEBUGLOG_ENABLE = 0
81   BM_SRV_DEBUGLOG_LEVEL  = 51
82   INTEGRITY_MONITOR      = 0
83   PREDEFINED_TRUSTED_UPDATER = 0
```



```
USR_DEBUGLOG_ENABLE      = 1
USR_DEBUGLOGLEVEL        = 273
SRV_DEBUGLOG_ENABLE      = 1
SRV_DEBUGLOGLEVEL        = 273
FW_USR_DEBUGLOG_ENABLE   = 1
FW_USR_DEBUGLOG_LEVEL    = 273
FW_SRV_DEBUGLOG_ENABLE   = 1
FW_SRV_DEBUGLOG_LEVEL    = 273
BM_SRV_DEBUGLOG_ENABLE   = 1
BM_SRV_DEBUGLOG_LEVEL    = 51
```

StellarProtect Legacy Mode installation fails

- In older versions of StellarProtect, the Support tool is different for SP and SPLM
- Collect the StellarProtect Legacy Mode logs using StellarSupport.exe
 - Modify StellarSetup.ini (see previous slide)
 - Re-run the installer to reproduce the issue, and do not close any installation windows when the installation fails
 - Execute StellarSupport.exe (in Agent Installer Package)
 - Click “Collect Debug Log”
 - Click “Open Folder” to get the archived zip file



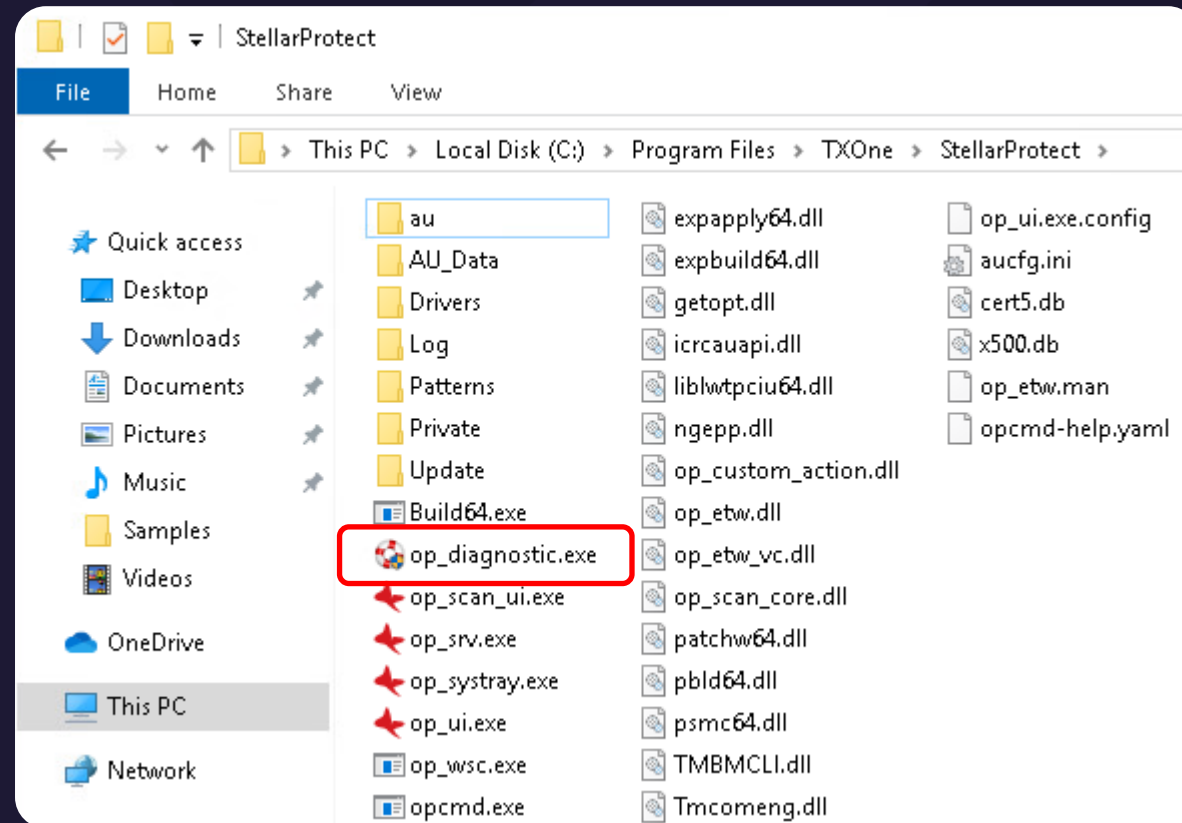
Scenario 7: Stellar Debug Log Collection Post Installation

Stellar v1.x ~ v3.0 SP1

Please notice some features in StellarOne are not available in the older version

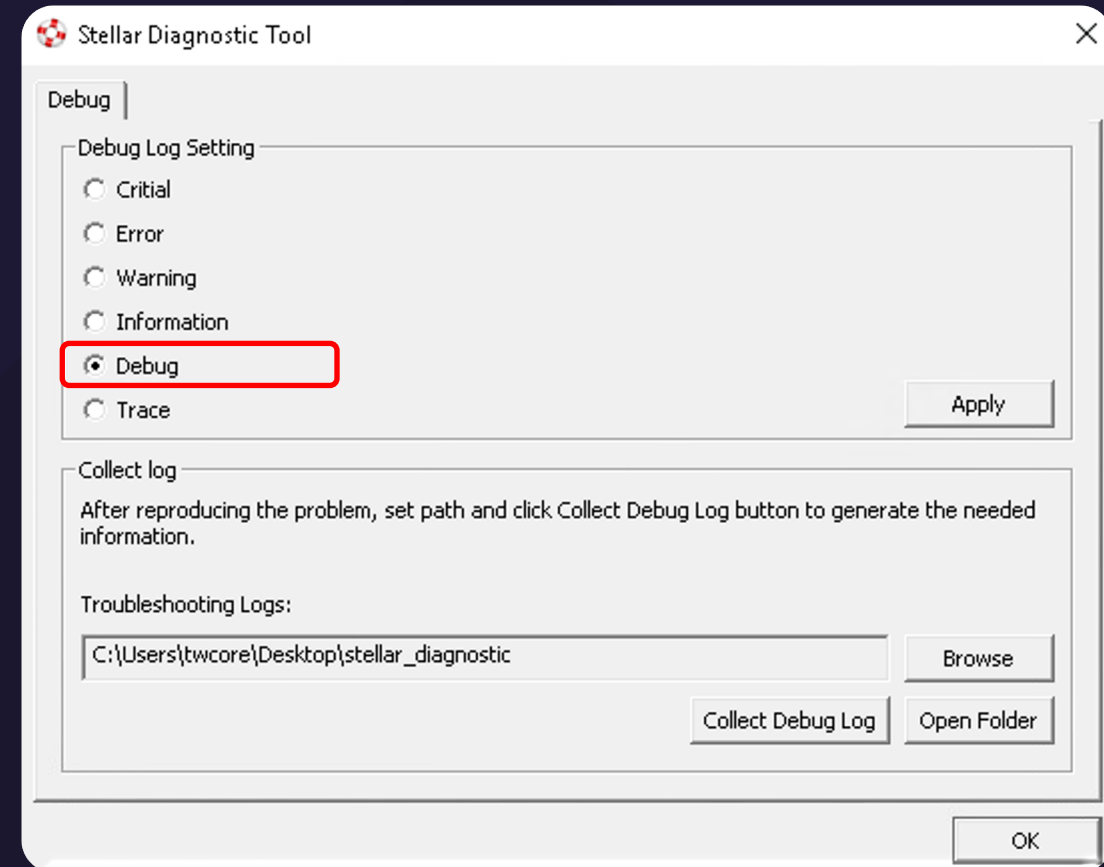
StellarProtect debug log from op_diagnostic.exe

- op_diagnostic is the built-in diagnostic tool for StellarProtect
- Path:<installation_dir>\op_diagnostic.exe



StellarProtect debug log from op_diagnostic.exe

- How to collect StellarProtect logs
 1. Select “Debug”
 2. Click “Apply”
 3. Reproduce the issue
 4. Click “Collect Debug Log”
 5. After the “Troubleshooting data collected” message box pop up, click “Open Folder” to get the archived zip file.
 6. Set debug level back to “Error”, click “Apply”



StellarProtect debug log from StellarOne

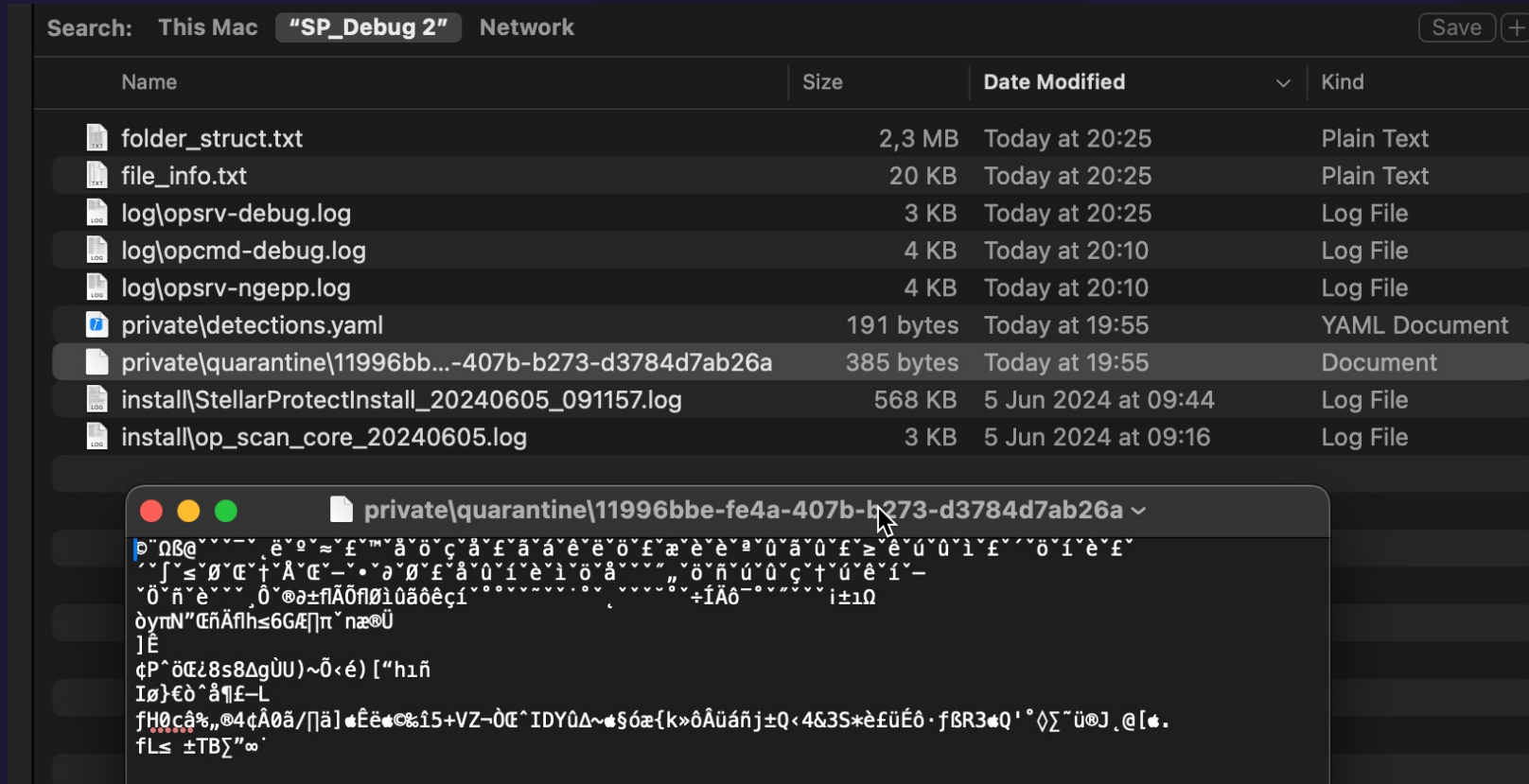
- Administrator could also collect StellarProtect debug log remotely via StellarOne management console
 - https://StellarOne_Server/rdqa

The screenshot displays the StellarOne management console interface. At the top, the 'stellarOne' logo is on the left, and the user 'admin (admin)' is on the right. Below the navigation bar, the 'DEBUG TOOL' section is active. The 'StellarProtect' endpoint is selected, indicated by a red box and a yellow circle with the number 1. The 'QQ-STELLAR' endpoint is highlighted with a red box and a yellow circle with the number 2. The 'Debug Logging' dropdown menu is open, showing 'Enable' and 'Disable' options, with a red box and a yellow circle with the number 3. The 'Collect Debug Information' button is highlighted with a red box and a yellow circle with the number 4. The table below shows the endpoint details:

Endpoint	Description	IP Address	Operating System	Last Connection ↓	Debug Logging Status
QQ-STELLAR		192.168.23.24	Windows 7 Professiona...	2023-07-03T12:05:48+...	● Debug logging disabled

Retrieve from remote quarantine

- The remote logs also contain the quarantine folder of the remote system



Retrieve a file from remote quarantine

1. go to https://YOUR_STELLARONE_URL/rdqa > StellarProtect >
select the endpoint from the list and click “collect debug Information”
2. that will give you a zip file with a lot of local logs of that endpoint,
but also with the encrypted quarantine file(-s)
3. setup a local VM and install a StellarProtect agent on it
4. copy the quarantine file (e.g. 66593088-6936-43b1-873a-5026a98cc95a)
to that local system e.g. in the %temp% folder

Retrieve a file from remote quarantine

5. from an elevated command prompt, run `c:\Program Files\TXOne\StellarProtect\opcmd -p service stop``

6. copy the quarantine file (the one from the ZIP) to `c:\Program Files\TXOne\StellarProtect\Private\Quarantine`

7. check the contents of the quarantine store: `c:\Program Files\TXOne\StellarProtect\opcmd.exe" -p quarantine show`

You should now see the file that you got from the remote system in your local quarantine store

```
C:\Windows\system32>"c:\Program Files\TXOne\StellarProtect\opcmd.exe" -p quarantine show
```

```
Password:
```

```
xxxxxxxx
```

Date Modified	Name
2024-06-28T19:55:21	11996bbe-fe4a-407b-b273-d3784d7ab26a

Original Location

C:\Users\txone\AppData\Local\Temp\TEMP1_~1.ZIP\samples\eicar.com.zip
--

Retrieve a file from remote quarantine

8. restart the service `opcmd -p service start`

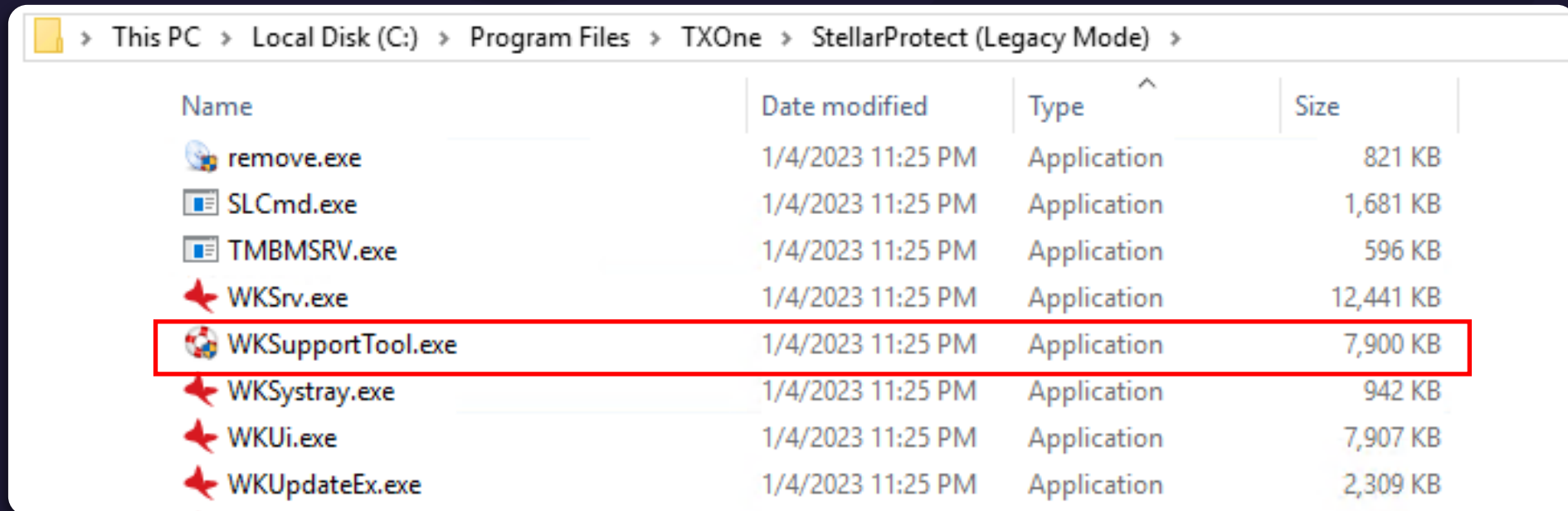
9. this is crucial: Locally recreate the original path where the file was found in, on the remote system (see previous screenshot).
That path must exist locally (!)

10. Make sure the system is in an isolated (forensics) environment

11 restore the file from quarantine `"c:\Program Files\TXOne\StellarProtect\opcmd.exe" -p quarantine restore 66593088-6936-43b1-873a-5026a98cc95a`

StellarProtect Legacy Mode debug log from op_diagnostic.exe

- The WKSupportTool is the built-in diagnostic tool for StellarProtect(Legacy Mode)
 - Path:<installation_dir>\WKSupportTool.exe



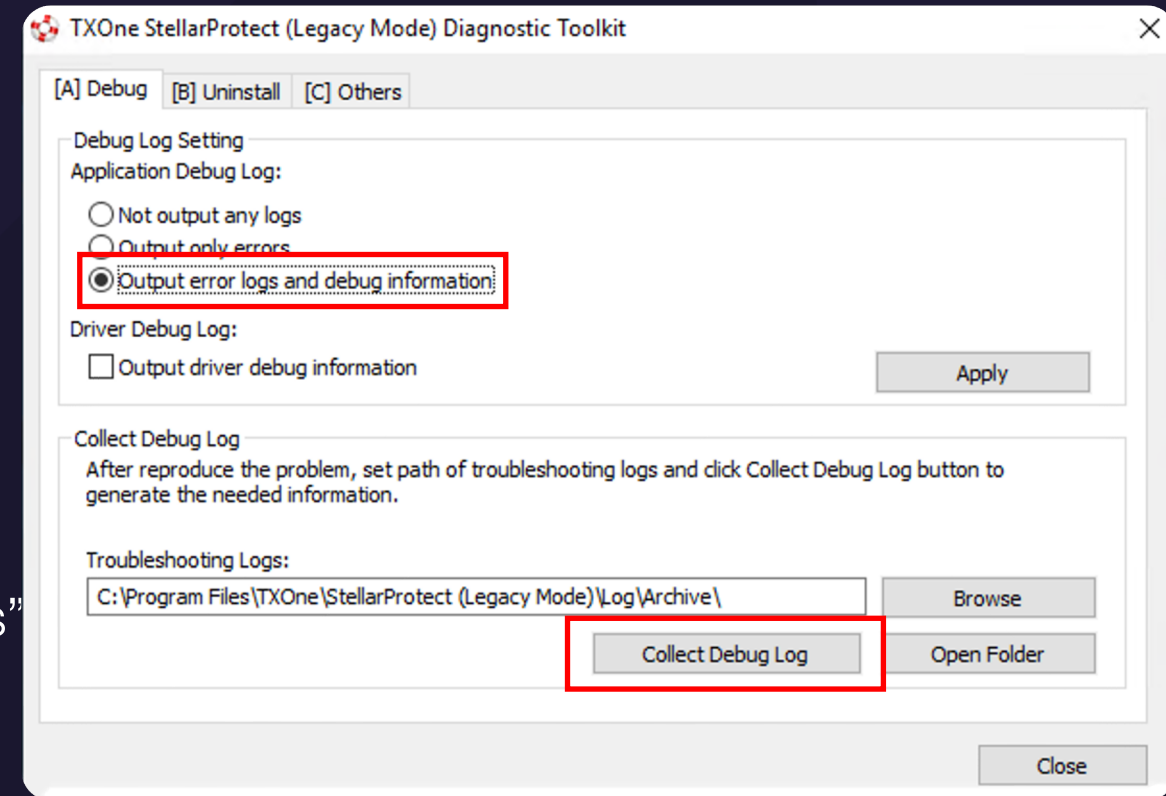
This screenshot shows a Windows File Explorer window with the address bar set to 'This PC > Local Disk (C:) > Program Files > TXOne > StellarProtect (Legacy Mode)'. The main area displays a list of files and folders. The file 'WKSupportTool.exe' is highlighted with a red rectangle. The list includes the following items:

Name	Date modified	Type	Size
remove.exe	1/4/2023 11:25 PM	Application	821 KB
SLCmd.exe	1/4/2023 11:25 PM	Application	1,681 KB
TMBMSRV.exe	1/4/2023 11:25 PM	Application	596 KB
WKSrv.exe	1/4/2023 11:25 PM	Application	12,441 KB
WKSupportTool.exe	1/4/2023 11:25 PM	Application	7,900 KB
WKSystray.exe	1/4/2023 11:25 PM	Application	942 KB
WKUi.exe	1/4/2023 11:25 PM	Application	7,907 KB
WKUpdateEx.exe	1/4/2023 11:25 PM	Application	2,309 KB

StellarProtect Legacy Mode debug log from op_diagnostic.exe

- How to collect StellarProtect(Legacy Mode) logs

1. Select “Output error logs and debug information”
2. (optional) Check “Output driver debug information”
3. Click “Apply”
4. Reproduce the issue
5. Click “Collect Debug Log”
6. After the “Troubleshooting data collected” message box pop up, click “Open Folder” to get the archived zip file.
7. Set debug level back to “Output only errors” click “Apply”



StellarProtect Legacy Mode debug log from StellarOne

- Administrator can also collect StellarEnforce debug log remotely via StellarOne management console
 - https://StellarOne_Server/rdqa

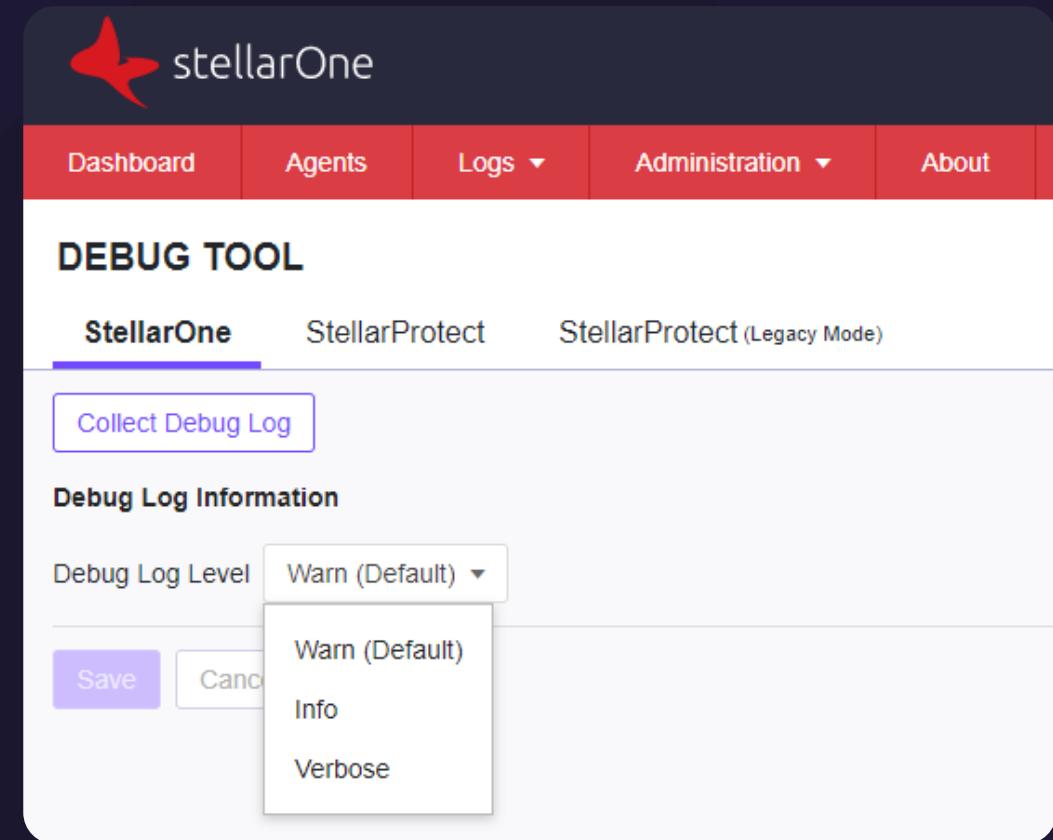
The screenshot displays the StellarOne management console interface. At the top, the 'stellarOne' logo is on the left, and the user 'admin (admin)' is on the right. Below the navigation bar, the 'DEBUG TOOL' section is active, showing tabs for 'StellarOne', 'StellarProtect', and 'StellarProtect (Legacy Mode)'. The 'StellarProtect (Legacy Mode)' tab is selected. Below the tabs, there are three buttons: 'Check Connection', 'Debug Logging', and 'Purge Collected Information'. The 'Debug Logging' button is highlighted with a red box and a yellow circle labeled '3'. A dropdown menu is open for 'Debug Logging', showing 'Enable' (selected) and 'Disable' options. The 'Enable' option is highlighted with a red box and a yellow circle labeled '4'. Below the dropdown, there is a 'Collect Debug Information' button, also highlighted with a red box and a yellow circle labeled '4'. On the left, the 'Endpoint' column of the table shows 'QQ-STELLARLM' selected, highlighted with a red box and a yellow circle labeled '2'. The table has columns for 'Endpoint', 'Description', 'IP Address', 'Operating System', 'Last Connection', and 'Debug Logging Status'. The first row shows 'QQ-STELLARLM' as the endpoint, with an IP address of '192.168.23.28' and an operating system of 'Windows 7 Professiona...'. The 'Last Connection' is '2023-07-03T11:55:37+...'. The 'Debug Logging Status' column has a 'Download' link. A yellow circle labeled '1' is placed above the 'StellarProtect (Legacy Mode)' tab. A yellow circle labeled '3' is placed above the 'Check Connection' button.

Endpoint	Description	IP Address	Operating System	Last Connection ↓	Debug Logging Status
QQ-STELLARLM	-	192.168.23.28	Windows 7 Professiona...	2023-07-03T11:55:37+...	Download

Scenario 8: StellarOne Troubleshooting Page (/rdqa)

StellarOne Troubleshooting Page (/rdqa)

- StellarOne Debug Log can be collected from https://StellarOne_Server/rdqa
 - Select the Debug Log Level: Verbose
 - Click on the “**Save**” button
 - Reproduce the issue (server and/or agent)
 - Click the “**Collect Debug Log**” button
 - Switch the debug log level back to Warn(Default)
 - Click on the “**Save**” button



Summary of troubleshooting tools

- **StellarSupport.exe** for troubleshooting when the installation fails.
Can be found in the in the Installer Package.
It is a wrapper around op_diagnostic.exe and WKSupportTool.exe
 - With StellarProtect: you can run this tool when installation fails
 - With StellarProtect LegacyMode: first change parameters in StellarSetup.ini in the installation folder). Then run this StellarSupport.exe tool
- **op_diagnostic.exe** is the built-in diagnostic tool for StellarProtect
Path:<installation_dir>\op_diagnostic.exe
- **WKSupportTool.exe** is the built-in diagnostic tool for StellarProtect Legacy Mode
Path:<installation_dir>\WKSupportTool.exe
- StellarOne troubleshooting:
 - See cli (e.g. env ls -> Status Fail -> No external disk detected
 - [https://\[stellarOne\]/rdqa](https://[stellarOne]/rdqa) -> Debug Log information
 - These logs are encoded in "journal" format (!)

Troubleshooting Security Inspection Solutions

Scenarios

1. The ElementOne unable to bootup
2. The ElementOne pattern update issue
3. The Portable Inspector cannot register to the ElementOne
4. Unable to access the ElementOne web UI
5. What is communication protocol between Portable and ElementOne
6. Storage capacity of PI quarantine files
7. Debug log collection

Scenario 1 - The ElementOne unable to boot up

Requirement

ElementOne



- Image corrupt
- VMWare
 - ESXi6
 - Workstation 14
- Manage up to 1,000 node
 - CPU: 4 cores
 - RAM: 4 GB

Forgot to create 2nd disk

ElementOne



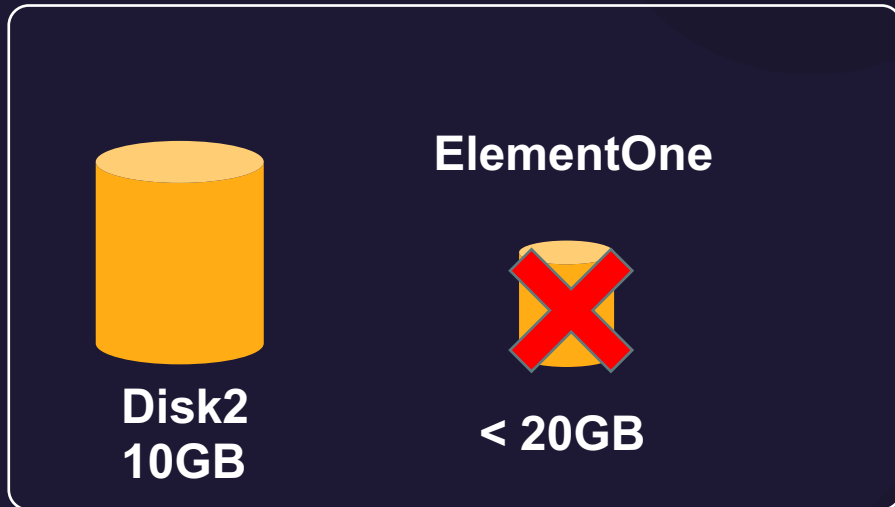
Disk1



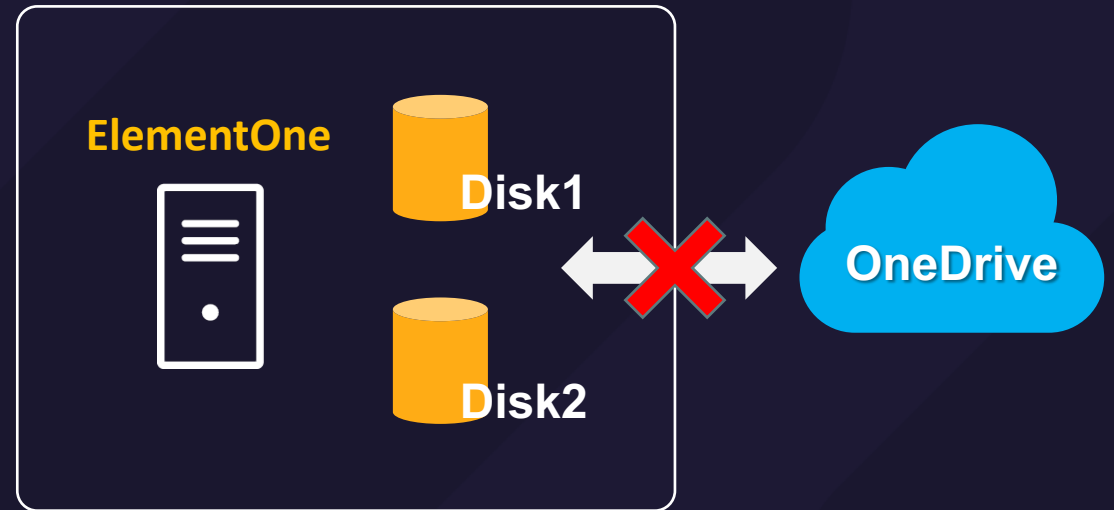
Disk2

Scenario 1 - The ElementOne not boot up

The size of the 2nd disk smaller than 20GB

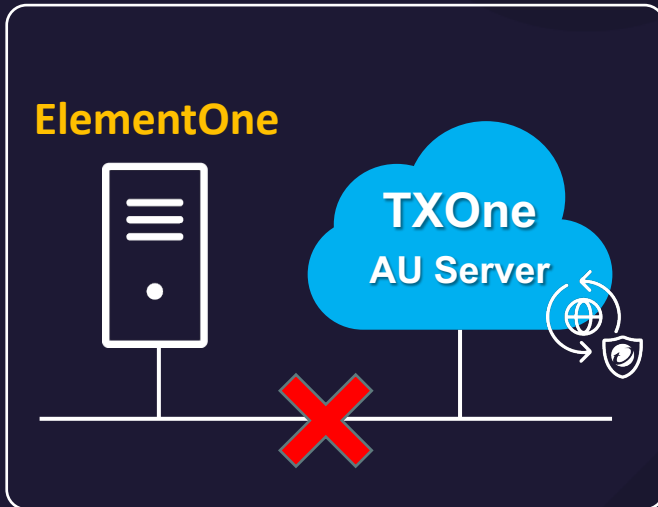


The Disk sync with Cloud drive services



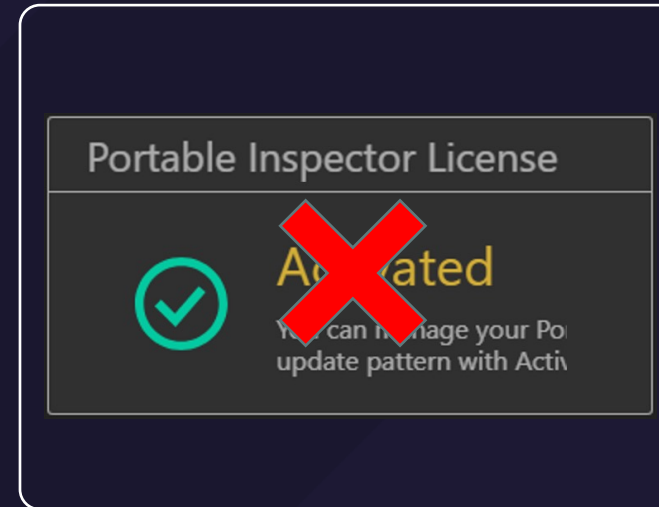
Scenario 2 - The ElementOne pattern update issue

Network connection issue



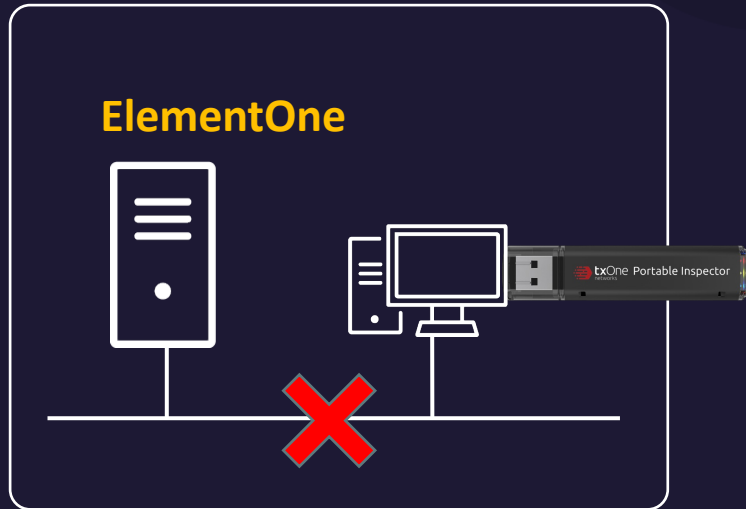
- The firewall deny the access to TXOne AU servers
- Network routing issue

There has no activated PI / PI Pro license

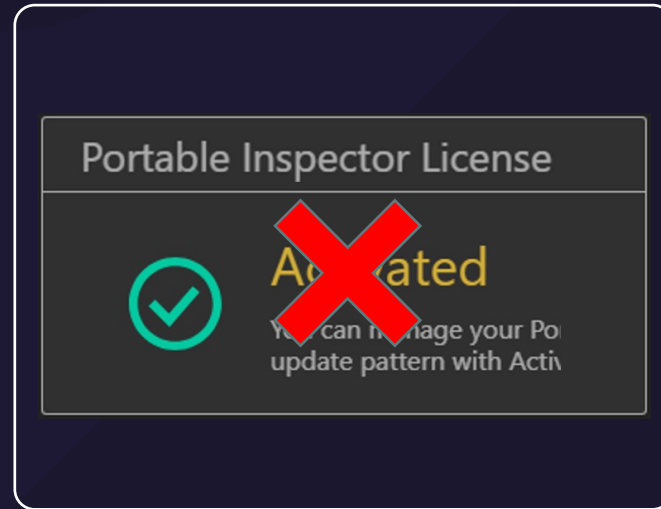


Scenario 3 - The Portable Inspector cannot register to the ElementOne

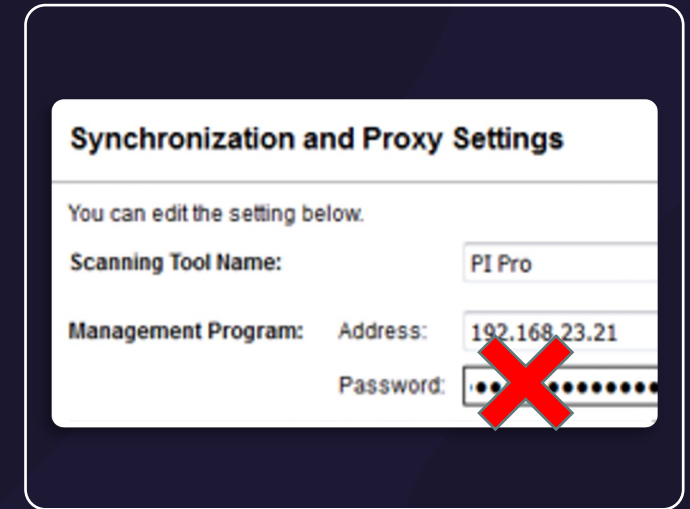
Network connection issue



There has no activated PI / PI Pro license

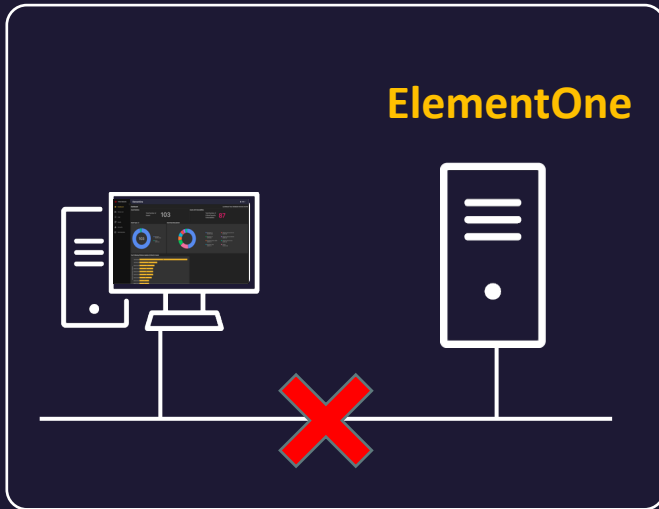


The wrong registration password



Scenario 4 - Unable to access the ElementOne web UI

Network connection issue



Browser issue



- The browser not use Chrome

The ElementOne time server not in sync



Scenario 5 – What is communication protocol between PI and E1



Registration : HTTPS

Operations : mTLS



Scenario 6 – Hidden partition - storage capacity of PI quarantine files



Pro edition : 33GB



Standard edition : 8GB

Types of data store in hidden partition

- Scan logs
- Asset information
- Quarantine files

It depends on if it contains quarantined files or not, in general, more than 10K Scan logs and Asset Info can be stored in a Scanning Tool.

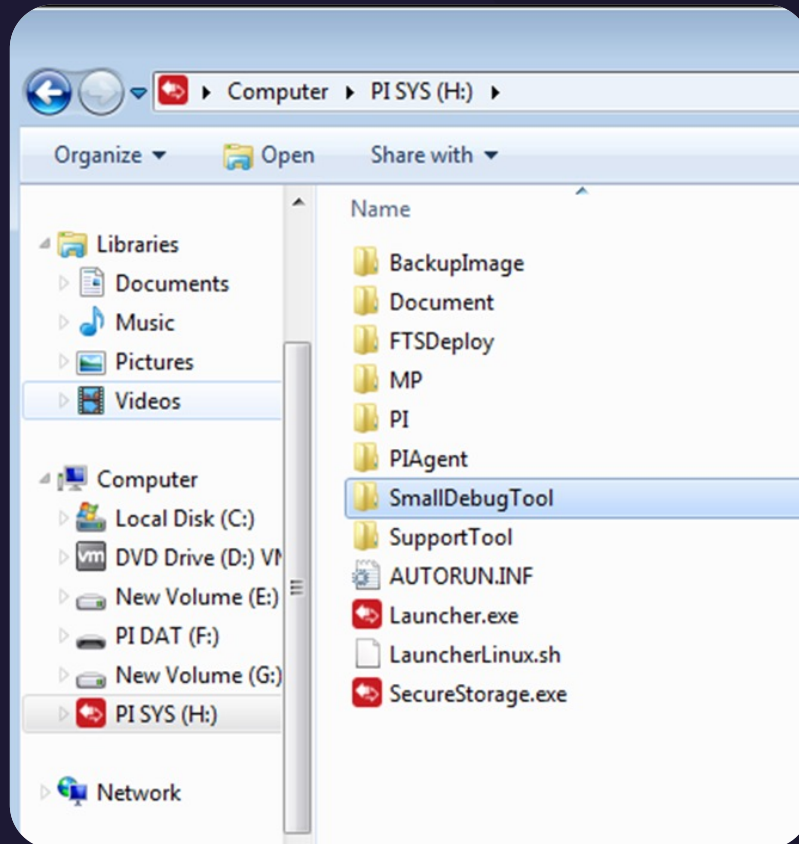
Troubleshooting ElementOne Debug log collection

Topics

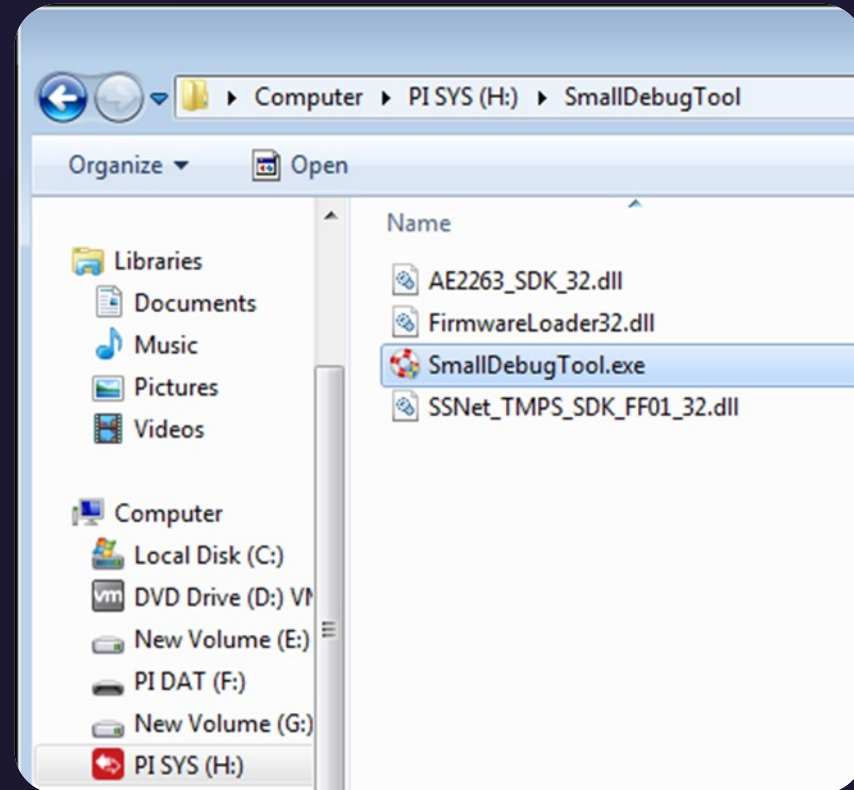
- How to collect the Scanning Tool / Secure Storage Debug log (Windows Platform)
 - Enabling the Scanning Tool debug log
 - Exporting logs from Scanning Tool
- How to collect the ElementOne debug log

Enabling the Scanning Tool debug log – Windows platform

- 1 Plug-in the Portable Inspector (Pro) Scanning Tool to the endpoint and navigate to the “PI SYS” drive.

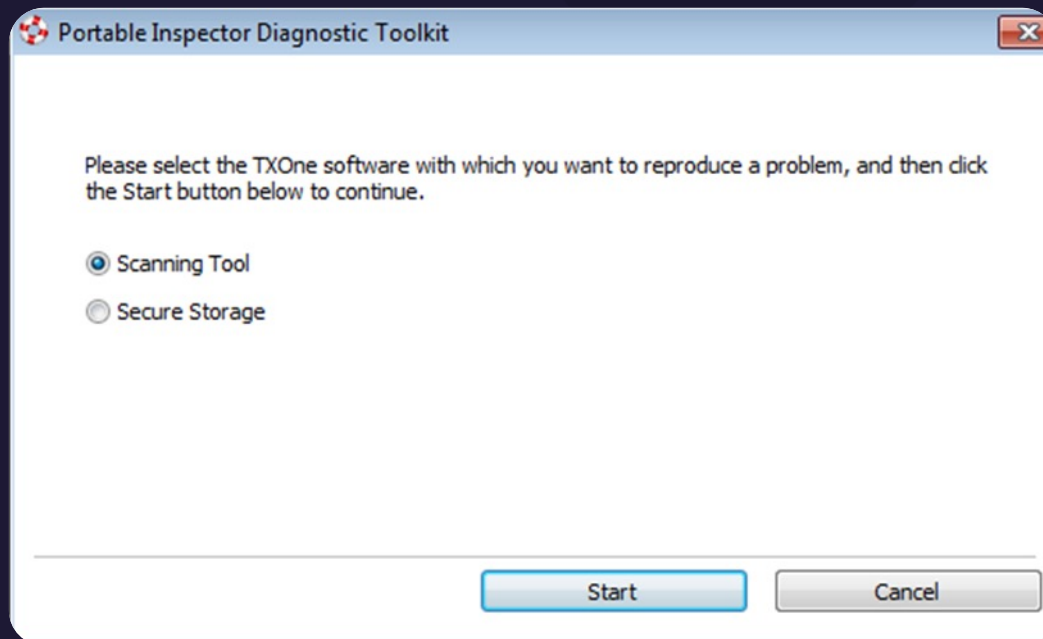


- 2 Launch the "SmallDebugTool.exe" found in the SmallDebugTool folder to collect logs.

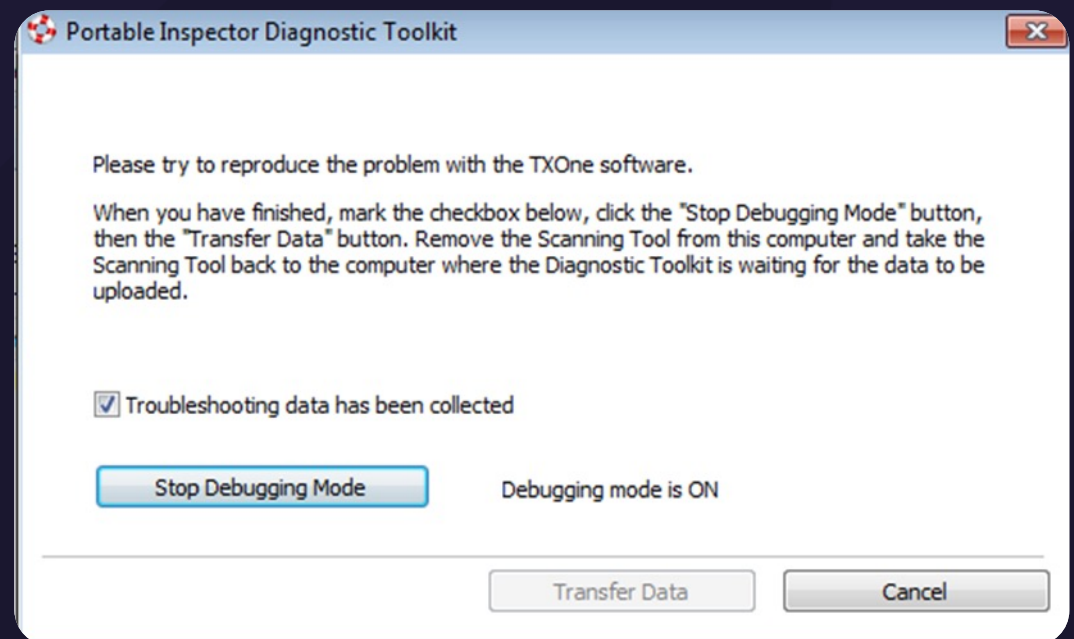


Enabling the Scanning Tool debug log – Windows platform

- 3** The debugging wizard will start automatically. Select Scanning Tool or Secure Storage and click the Start button.

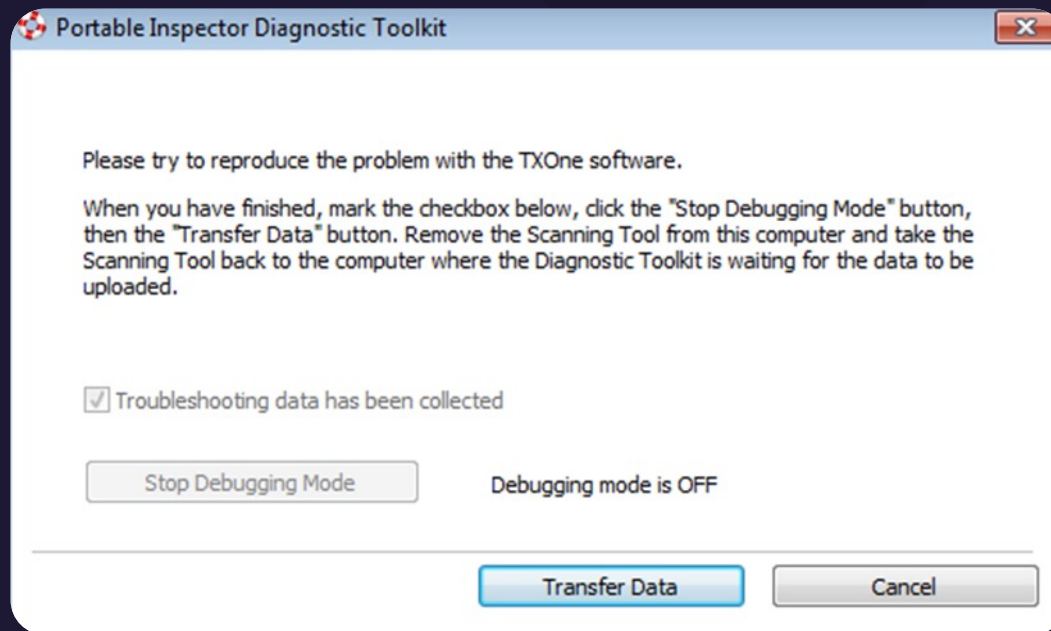


- 4** Reproduce the issue with the scanning tool. When the issue is reproduced, select “Troubleshooting data has been collected” and click “Stop Debugging Mode”.

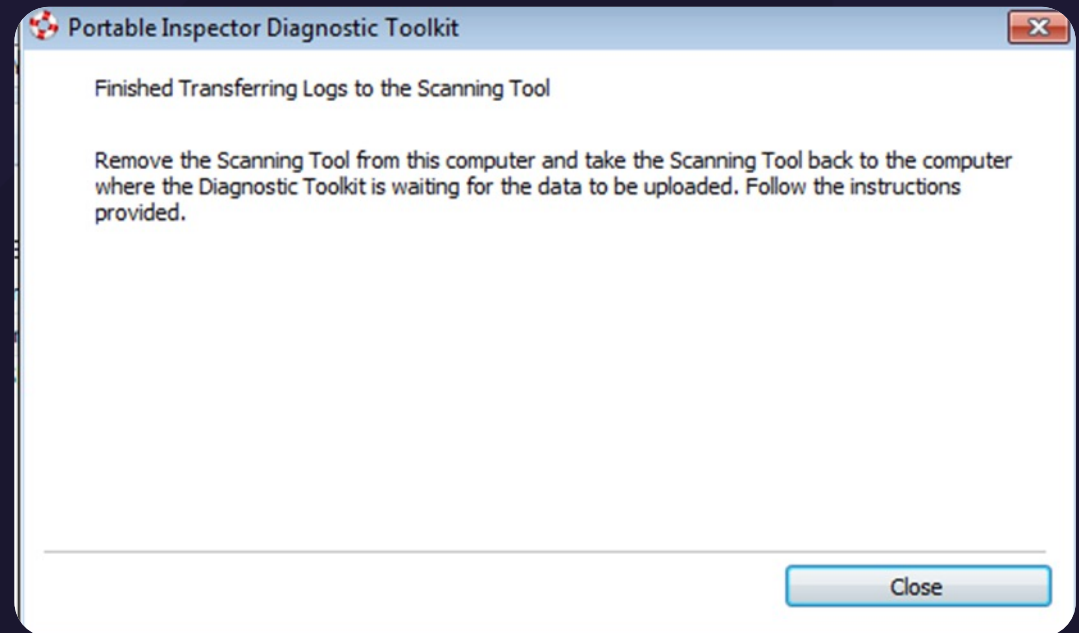


Enabling the Scanning Tool debug log – Windows platform

- 5** After data is done saving, click "Transfer Data"

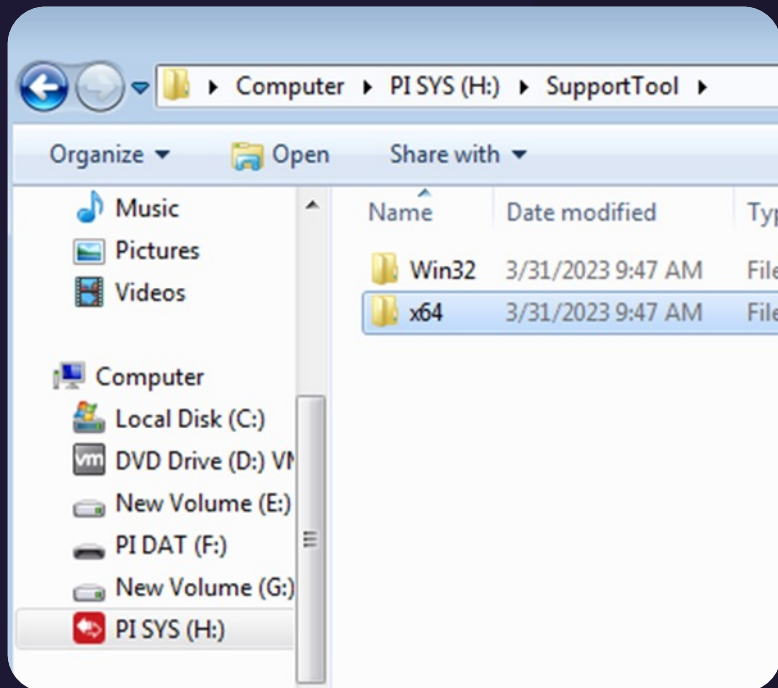


- 6** After the debug log is collected, click "Close".

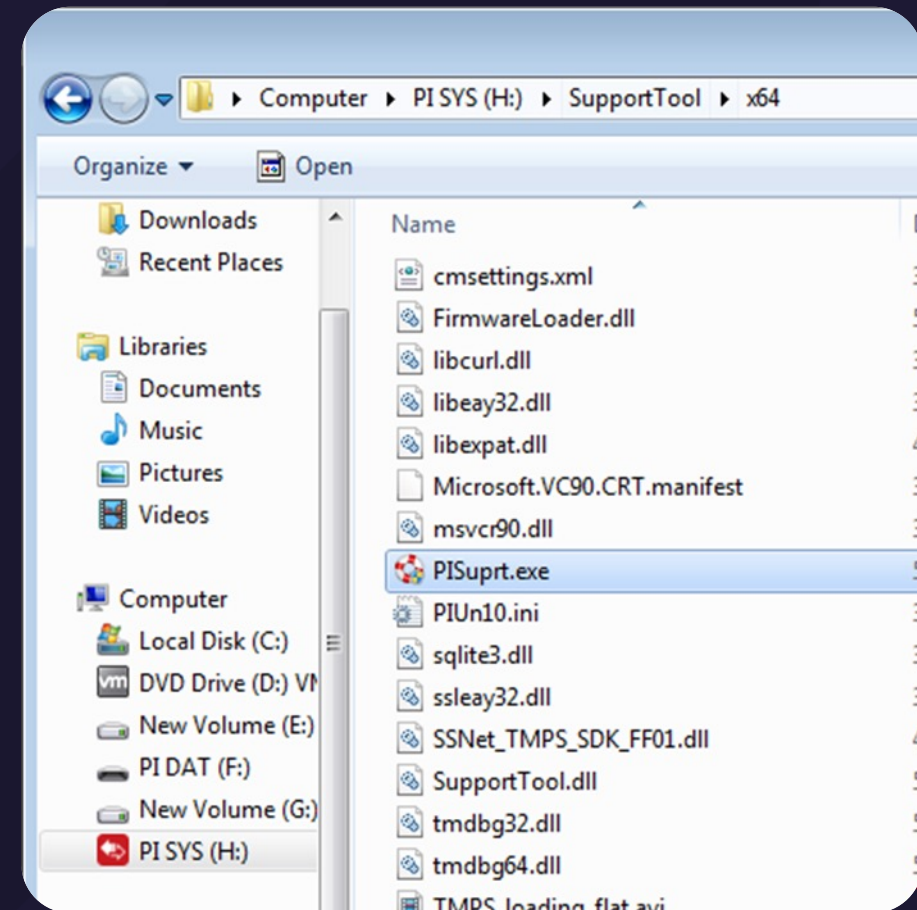


Exporting logs from Scanning Tool

- 1 Plug-in the PI Pro. Navigate to the “PISYS” drive and access the “SupportTool” folder.

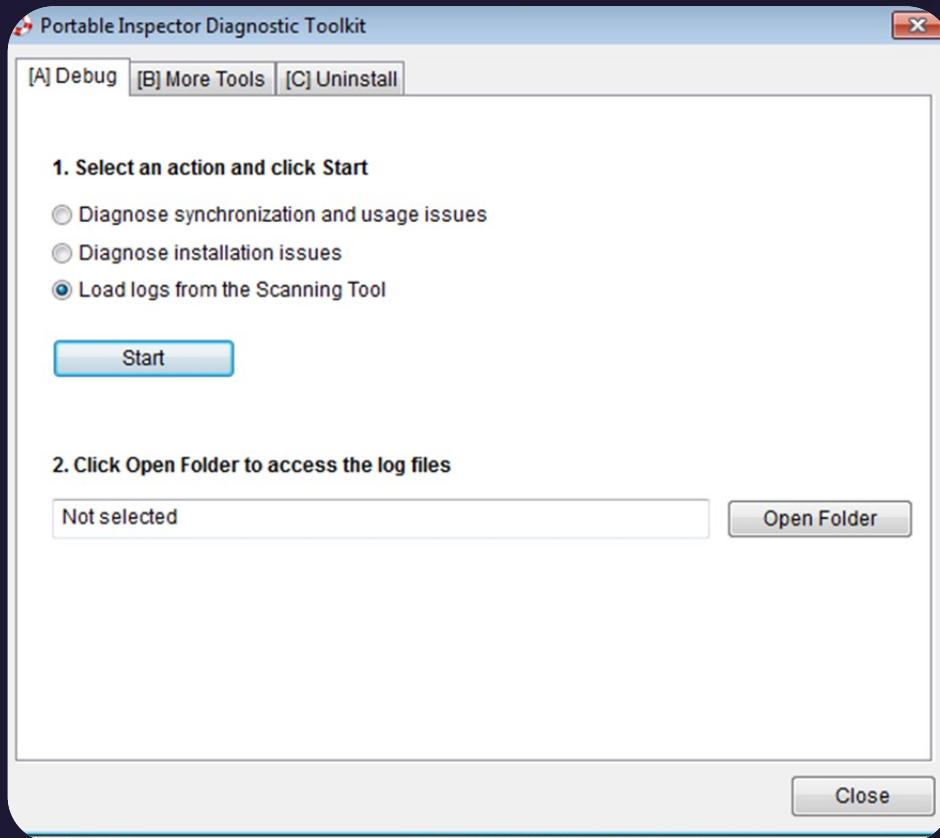


- 2 Choose the directory appropriate for your environment and execute the “PISuprt.exe”.

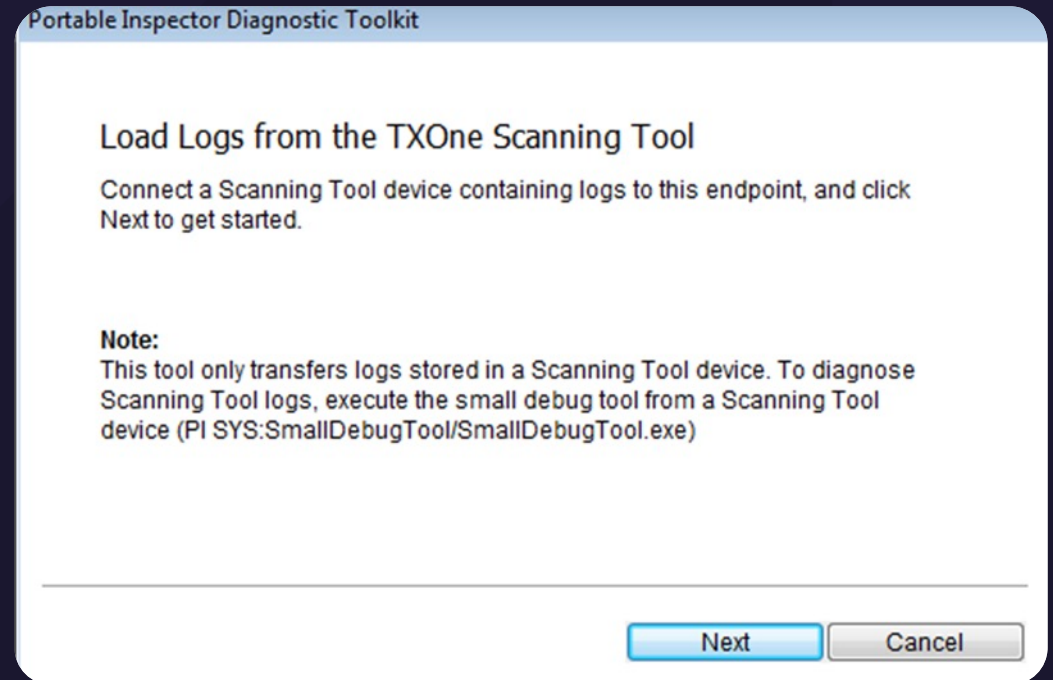


Exporting logs from Scanning Tool

- 3** In the Debug tab, select Load logs from the Scanning Tool and click Start.

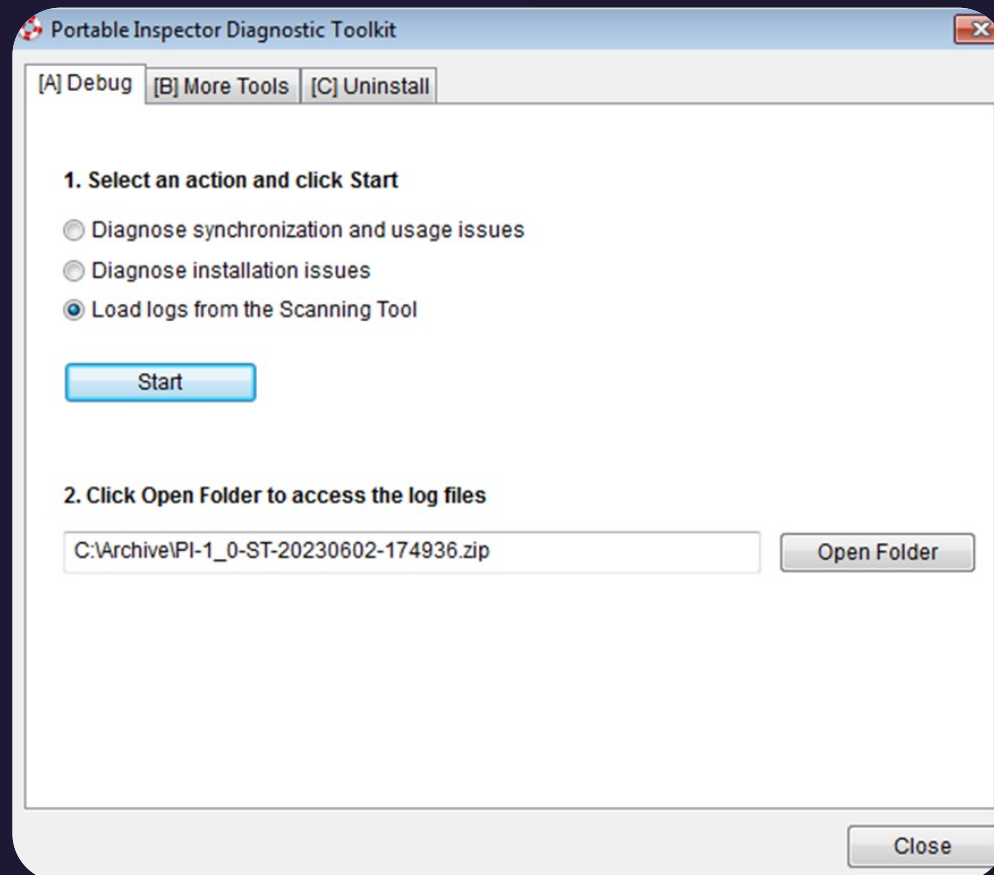


- 4** Connect the Scanning Tool to the endpoint and click Next. The storage path of the logs are indicated.



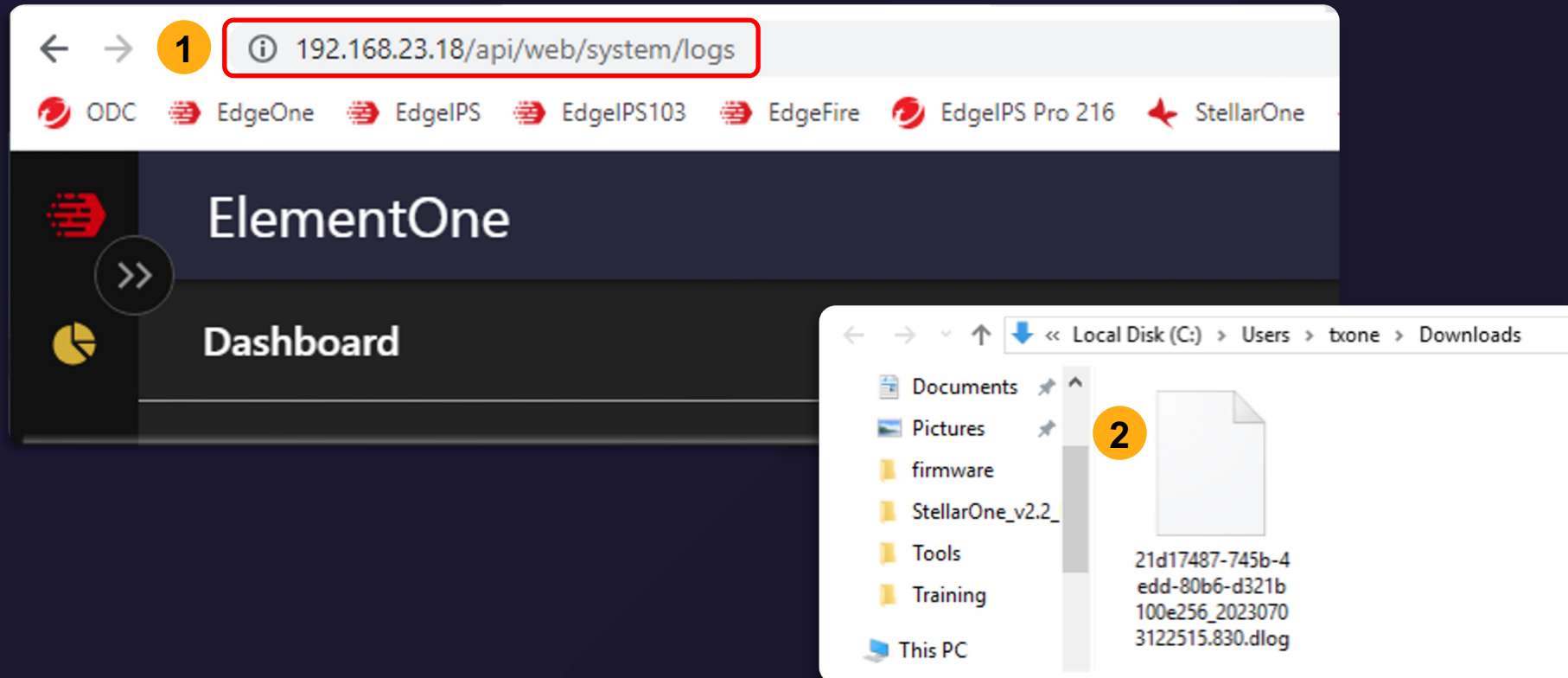
Exporting logs from Scanning Tool

- 5** Click Open Folder to navigate to the path. Locate and open the zip file to verify that the debug logs have been successfully generated.



Debug log collection from ElementOne web UI

- https://ElementOne_web_UI_IP/api/web/system/logs





Thank you

Let's Keep The Business Growing